
APPENDIX E-2

ENVIRONMENTAL SAMPLING PLAN FOR SOIL, GROUNDWATER, AND SURFACE WATER QUALITY MONITORING FOR THE AMMONIUM PERCHLORATE ROCKET MOTOR DESTRUCTION FACILITY

ENVIRONMENTAL SAMPLING PLAN SOIL, GROUNDWATER AND SURFACE WATER QUALITY MONITORING

**for
The Ammonium Perchlorate Rocket Motor Destruction
(ARMD) Facility**

EPA ID No. PA6213820503

**Submitted to
The Pennsylvania Department of Environmental Protection**

**February 2015
Revised May 2016**

Prepared for:

Letterkenny Army Depot
Chambersburg, Pennsylvania



Prepared by
Redhorse Corporation
San Diego, California 92101
(619) 722-2282

El Dorado Engineering, Inc.
2964 W. 4700 South, Suite 109
Salt Lake City, UT 84129
(801) 966-8288

**Environmental Sampling Plan
Soil, Groundwater and Surface Water Quality Monitoring
For
The Ammonium Perchlorate Rocket Motor Destruction (ARMD) Facility**

EPA ID No. PA6213820503

ARMD ESP, Letterkenny Army Depot

CONTRACTOR Program

LEAD Environmental: _____ Date: _____

DISTRIBUTION LIST

Name	Responsibility	Affiliation

CONTENTS

DISTRIBUTION LIST	ii
1.0 PROJECT BACKGROUND AND PURPOSE	1
2.0 PRINCIPAL DECISION-MAKERS	4
3.0 LEAD SITE DESCRIPTION AND PHYSICAL SETTING.....	4
4.0 DATA QUALITY OBJECTIVES AND CRITERIA.....	7
4.1 DATA QUALITY OBJECTIVES.....	7
4.2 MEASUREMENT QUALITY OBJECTIVES	7
5.0 MONITORING OF THE ARMD TTC SUMP AND WITNESS ZONE.....	9
6.0 INVESTIGATION OF SOIL, SEDIMENT, SURFACE WATER AND GROUNDWATER	10
6.1 APPROACH	10
6.2 TECHNICAL OR REGULATORY STANDARDS.....	16
6.3 PROJECT TIMELINE	20
6.4 ANALYTICAL METHODS AND REPORTING LIMITS	21
6.5 FIELD DOCUMENTATION	24
6.6 SAMPLE COLLECTION	24
6.6.1 Soil Samples.....	30
6.6.2 Stormwater Runoff Samples – Water and Sediment	32
6.6.3 Surface Water Samples	34
6.6.4 Groundwater Elevation	36
6.6.5 Groundwater Samples	37
6.6.6 Collection of Quality Assurance and Quality Control Samples	40
6.7 SAMPLING EQUIPMENT DECONTAMINATION PROCEDURES	42
6.8 SAMPLE HANDLING AND CUSTODY.....	43
6.8.1 Sample Identification	43
6.8.2 Sample Labels.....	44
6.8.3 Sample Documentation	44
6.8.4 Chain-of-Custody.....	44
6.8.5 Sample Shipment	46
6.9 EQUIPMENT TESTING, INSPECTION, AND MAINTENANCE	46
6.9.1 Calibrating Field Equipment.....	47

6.9.2	Maintaining Field Equipment	47
6.9.3	Inspection and Acceptance of Supplies and Consumables	47
7.0	QUALITY ASSURANCE AND QUALITY CONTROL.....	48
7.1	QUALITY ASSURANCE AND QUALITY CONTROL DEFINITIONS	49
7.1.1	Precision.....	49
7.1.2	Accuracy	49
7.1.3	Representativeness	50
7.1.4	Completeness	50
7.1.5	Comparability	50
7.1.6	Method Detection and Quantitation Limits	50
7.2	QUALITY ASSURANCE AND QUALITY CONTROL FIELD SAMPLES	51
7.2.1	Temperature Blanks	51
7.2.2	Trip blanks	51
7.2.3	Duplicates	51
7.2.4	Rinsate Blanks	52
7.2.5	Source-Water blanks	52
7.3	LABORATORY QUALITY ASSURANCE AND QUALITY CONTROL	52
8.0	SPECIAL TRAINING AND CERTIFICATION	53
8.1	HEALTH AND SAFETY TRAINING	53
8.2	SUBCONTRACTOR TRAINING	54
8.3	SPECIALIZED TRAINING AND CERTIFICATION REQUIREMENTS.....	54
9.0	DATA MANAGEMENT.....	54
10.0	DATA ANALYSIS AND REPORTING.....	55
	REFERENCES	57

FIGURES

FIGURE 1 LEAD Location Map	2
FIGURE 2 Preliminary Site Plan of the ARMD.....	3
FIGURE 3 ARMD Sample Collection Locations.....	12

TABLES

TABLE 1 Data Quality Objectives	8
TABLE 2 Proposed Soil, Surface Water, Groundwater, Stormwater Runoff, and Runoff Sediment Sample Location and Sampling Rationale.....	14
TABLE 3 40 CFR Part 264.94 Maximum Concentration of Constituents for Groundwater Protection	17
TABLE 4 Pennsylvania and U.S. Environmental Protection Agency Standards and Guidelines	18
TABLE 5 Implementation Schedule for Sampling, Analysis, and Reporting.....	20
TABLE 6 Samples Volumes, Containers, Preservatives, and Holding Times	23
TABLE 7 Proposed Soil, Surface Water, Groundwater, Stormwater Runoff, and Runoff Sediment Sample Field Sampling Identification and Collection.....	25
TABLE 8 Quality Control Samples for Precision and Accuracy	48

APPENDICES

APPENDIX A Volatile Organic Compound List

ATTACHMENTS

ATTACHMENT A PADEP Well Installation Plan Approval Letter & Weston Solutions, Inc.
Well Installation Plan Letter Report

ATTACHMENT B Depot-Wide Analytical Methods And Reporting Limits

ACRONYMS AND ABBREVIATIONS

%	Percent
±	Plus or minus
Ag	Silver
Al	Aluminum
ANOVA	Analysis of variance
ARMD	Ammonium Perchlorate Rocket Motor Destruction Facility
As	Arsenic
Ba	Barium
bgs	Below ground surface
Bi	Bismuth
°C	Degrees Celsius
CASRN	Chemical Abstract Services Registry Number
Cd	Cadmium
CERCLA	Comprehensive Environmental Response, Compensation, and Liability Act
CFR	Code of Federal Regulations
Cl	Chloride
ClO ₄	Perchlorate
COC	Chain of custody
CPR	Cardiopulmonary resuscitation
Cr	Chromium
Cu	Copper
CZM	Coastal Zone Management
DQO	Data quality objective
DTW	Depth to water
EDD	Electronic data deliverable
EDE	El Dorado Engineering
EPA	U.S. Environmental Protection Agency
EQL	Estimated Quantitation Limit
ESP	Environmental Sampling Plan
F	Fluoride
Fe	Iron
FTL	Field team leader
GC	Gas chromatography
GC/MS	Gas Chromatography Mass Spectrometry
HASP	Health and safety plan
Hg	Mercury
ID	Identification
ICP	inductively coupled plasma
IRDMIS	Installation Restoration Data Management Information System
K	Potassium

LCS	Laboratory control sample
LEAD	Letterkenny Army Depot
Mangi	Mangi Environmental Group
MCL	Maximum contaminant level
MCLG	Maximum contaminant level goal
MDL	Method detection limit
mg/kg	Milligram per kilogram
mg/L	Milligram per liter
ml	Milliliter
Mo	Molybdenum
MQO	Measurement quality objective
MS	Matrix spike
MSC	Medium-Specific Concentration
MSD	Matrix spike duplicate
mV	Millivolt
NO ₂ /NO ₃	Nitrates/Nitrites
NTU	Nephelometric Turbidity Unit
ORP	Oxidation reduction potential
OSHA	Occupational Safety and Health Administration
oz	Ounce
P	Phosphorus
Pa. Code	Pennsylvania Code
PADEP	Pennsylvania Department of Environmental Protection
PARCC	Precision, accuracy, representativeness, completeness, and comparability
PAS	Pollution abatement system
Pb	Lead
pH	Measurement of acidity or basicity
PPE	Personal protective equipment
PQL	Practical Quantitation Limit
QA	Quality assurance
QAPP	Quality assurance project plan
QA/QC	Quality assurance/quality control
QC	Quality control
RCRA	Resource Conservation and Recovery Act
Redhorse	Redhorse Corporation
RL	Reporting limit
RPD	Relative percent difference
S/SO ₄	Sulfur/sulfate
Se	Selenium
SiO ₂	Silica (also known as silicon dioxide)
Sn	Tin
SOP	Standard operating procedure

SSL	Soil Screening Level
TBD	To be determined
TOC	Top of casing
TTC	Thermal Treatment Chamber
µg/L	Micrograms per liter
µm	Micron
USDA	U.S. Department of Agriculture
VOA	Volatile organic analysis
VOC	Volatile organic compound
Weston	Weston Solutions, Inc.
Zr	Zirconium

1.0 PROJECT BACKGROUND AND PURPOSE

Letterkenny Army Depot (LEAD), located near Chambersburg, Pennsylvania (Figure 1), is a United States Government-Owned, Government-Operated, facility that provides supply and maintenance support to the United States Armed Forces. LEAD is under the command structure of the U.S. Army Aviation and Missile Lifecycle Management Command.

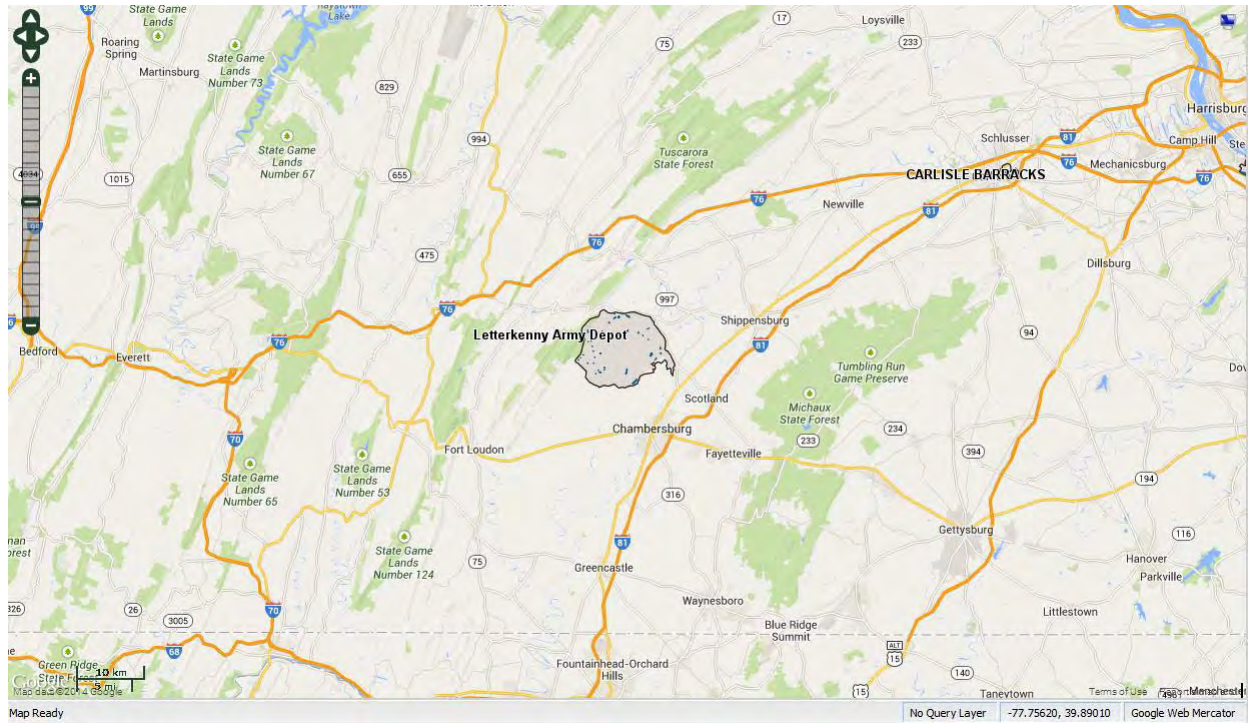
LEAD overhauls, rebuilds, modifies, tests, and repairs a wide variety of missile systems ranging from air-launched missiles to large ground-based air defense missile systems. LEAD also overhauls, modifies, and converts various ground support vehicles. In addition to equipment maintenance, LEAD also receives, stores, issues, renovates, and demilitarizes conventional ammunition.

LEAD is permitting and building a new destruction facility. The Ammonium Perchlorate Rocket Motor Destruction Facility (ARMD) will use confined burning with emissions control to treat solid propellant rocket motors that are currently treated at the Open Burning Grounds at LEAD. The ARMD includes static firing of full motors and containing and capturing the exhaust for treatment in a pollution abatement system (PAS). A confined burn system will utilize an ignition source to initiate combustion (rather than an auxiliary burner). A preliminary site plan of the ARMD is presented in Figure 2.

This Environmental Sampling Plan (ESP) outlines sampling activities to be conducted in conjunction with the new ARMD. Sampling activities will be initiated in response to requests from the Pennsylvania Department of Environmental Protection (PADEP) to establish baseline conditions at the site. Up to four, but a minimum of two baseline sampling events should be completed prior to start up of the ARMD to provide sufficient data for statistical analysis. Subsequent annual sampling requested by PADEP will be used to satisfy regulatory requirements for the protection of soil, surface water, and groundwater quality. Implementation of this ESP will establish baseline conditions and subsequent annual sampling will determine if the ARMD operation is adversely impacting the surrounding environment. The baseline and subsequent annual monitoring results will be submitted to PADEP, in addition to groundwater contour maps.

This ESP consists of a field sampling plan and quality assurance/quality control (QA/QC) approaches in a single document, to guide the field, laboratory, and data-reporting efforts associated with this soil, groundwater, and surface water monitoring project. This ESP addresses quality assurance elements to facilitate collection, analysis, and reporting of valid, precise, and accurate data. A depot-wide QAPP for environmental services supporting Installation Restoration Program investigations by contractors at LEAD also provides additional information on analytical methods and reporting limits (Weston 2003).

FIGURE 1
LEAD LOCATION MAP



2.0 PRINCIPAL DECISION-MAKERS

Principal decision makers involved in the soil, groundwater, and surface water monitoring activities at the ARMD include LEAD, the U.S Army Corp of Engineers and PADEP. Until such time as the group contracted to perform sampling and investigation efforts is known, this ESP references that entity as “CONTRACTOR.”

3.0 LEAD SITE DESCRIPTION AND PHYSICAL SETTING

Site Description and Topography

Surface elevations throughout the LEAD property range from approximately 600 to 800 feet above mean sea level, except for the northwest portion the Depot, where the elevation increases abruptly to more than 2,300 feet above sea level in the vicinity of Broad Mountain. A portion of the LEAD facility site includes 2,900 acres of mountainous wooded land along Blue or North Mountain with elevations ranging from 700 feet to 2,300 feet above sea level. Slopes rising in excess of 40 feet per 100 feet are found in the mountainous areas (Mangi, 2010).

The ARMD site is in a remote area of LEAD among uniquely compatible land uses such as missile maintenance, ammunition storage, and the Open Burning and Open Detonation grounds. Land use around LEAD is commercial except for a tract of agricultural land adjacent to the south side of the depot. Forestland (Buchanan State Forest) is about 4,000 feet west of the ARMD site, adjacent to the western boundary of LEAD. The structures at the ARMD site include the confined burn unit and two buildings that will be used for processing and de-segmenting the rocket motors. There are no water wells or injection wells in the ARMD area (Mangi, 2010).

Geology

The LEAD facility is located in the Cumberland Valley portion of the Great Valley in the eastern most section of the Ridge and Valley Province of the Appalachian Mountains. LEAD straddles two major geologic structural features; the South Mountain Anticlinorium to the east and the Massanutten Synclinorium to the west. The eastern section of LEAD is underlain primarily by carbonate rocks (limestones and dolomites) and is part of the South Mountain Anticlinorium (Mangi, 2010).

The Martin & Martin Inc. surface geology map presented in the RCRA Part B permit application depicts Chambersburg Formation, Martinsburg Formation, and St. Paul Group in the ARMD area of LEAD. The Martinsburg Formation, ranging in thickness from several feet (near fault contacts) to an estimated 1,000 feet, consists predominantly of black carbonaceous shale interbedded with sandstone and siltstone. The Chambersburg Formation is a dark-gray, argillaceous, cobbly limestone with abundant irregular shaley partings, and the St. Paul Group consists of a micritic and fossiliferous limestone with some black chert and dolomite (Redhorse/EDE 2014).

The bedrock formation in the ARMD area may provide a pathway to nearby agricultural activity. According to LEAD, the nearest downgradient groundwater consumers are between 1.5 and 2 miles southwest of the ARMD. This area is a mix of public water and groundwater consumers (LEAD, 2016). Further research into identification of water users and water supplies, if needed, will be addressed outside this ESP.

Soil

Based on the NRCS Soil Survey for Franklin County, there are three dominant soil series types found at the ARMD site area within LEAD: Weikert, Berks, and Beddinton. These soils are characterized as shallow to deep and well-drained. These acidic soils are weathered from shale, siltstone and acid sandstone and are associated with valley bottoms. Also near the ARMD site is Swanpond-Edom complex soil (USDA Natural Resources Conservation Service Soil Survey), which is residuum weathered from limestone and from limestone and shale. (Mangi, 2010).

Watersheds

Surface water drainage at LEAD is divided into two watersheds: the Susquehanna River to the northeast and the Potomac River to the southwest. Both the Susquehanna and the Potomac eventually drain into the Chesapeake Bay. Because LEAD is in the Chesapeake Bay region, it is subject to the 1987 Chesapeake Bay Agreement. Surface water runoff from the northeast portions of the LEAD facility is all contained in the Susquehanna watershed. Surface water runoff from the southwest portion of LEAD discharges to either Dennis Creek, Back Creek, Rocky Spring Branch, or Conococheague Creek, all contained in the Potomac River watershed. Drainages near the ARMD site area are ephemeral or intermittent with the stream channels carrying water only in winter and spring or after heavy rains. Surface water generally flows from northwest to southeast across the region surrounding the ARMD (Mangi, 2010). However in the ARMD area surface water generally flows from the northeast towards the southwest.

Groundwater

There is currently no demand for groundwater at LEAD and no such future demand is anticipated. Water is currently supplied to LEAD by surface water storage from Letterkenny Reservoir, which is located a few miles north of LEAD. LEAD is largely underlain by shales although limestone does occur in some drainages and in a narrow belt along the base of Broad Mountain. The Martinsburg Formation is generally a good aquifer yielding water of decent quality, although high iron and manganese concentrations may occur. Hydrogen sulfide gas is occasionally found and degrades water quality. Water in this aquifer is hard and unsuitable for some industrial processes. There is a close relationship between well yield and topography. Wells located in the area of low topographic expression have significantly greater yields than wells on upland locations. Geologically, wells located along fracture traces also have higher yields (Mangi, 2010).

Wetlands and Floodplains

Eight wetland types are present at LEAD: Lacustrine, Palustrine Aquatic Bottom, Palustrine Emergent, Palustrine Forested, Palustrine Open Water, Palustrine Scrub-shrub, Palustrine Unconsolidated Bottom, and Riverine. Vernal pools exist on portions of LEAD, with one pool located near the ARMD site, although the construction site itself will be located on previously cleared land (Mangi, 2010).

LEAD does not lie on any significant floodplain and is above the 100-year flood level of the Conococheague Creek. The result is that serious flooding is not a threat to operations at LEAD (Mangi, 2010).

All LEAD property is located within the Commonwealth of Pennsylvania Coastal Zone Management (CZM) Program boundary. Established by an Executive Order and approved in 1978, the CZM Program is a network of state laws and policies designated to protect coastal and marine resources. This includes the Chesapeake Bay, into which water from the streams and tributaries on LEAD eventually flow.

Vegetation

There are approximately 6,264 acres of forest land on LEAD. Three forest vegetation communities exist on LEAD; deciduous forests, coniferous dominated forests, and mixed coniferous and deciduous forests. Dominant tree species on the property include red oak, black oak, white oak, chestnut oak, yellow poplar, white ash, hickories, and red maple.

The majority of the terrestrial habitat on LEAD consists of open fields and second or third growth forest. Of the total 18,281 acres on LEAD, approximately 35 percent is forested and 52 percent is open fields, 1 percent is water, and the remaining 12 percent is mostly developed with scattered vegetation. Existing forested and wooded stands represent second and third-growth forests. Non-forested areas are principally old-field type successional areas, dominated by grass species.

There are approximately 9,250 acres of open habitat on the LEAD property, consisting principally of grassland fields that are incorporated into the agricultural leasing program at the Depot. Open areas also include buffer strips (roadways, areas surrounding igloos, field borders) that serve as fire breaks, edge habitat, and buffers for the protection of water resources and sensitive habitat (Mangi, 2010).

The ARMD site vicinity consists mainly of young second or third-growth mixed oak-hickory forest, some agricultural land, open areas, and several very small wetland patches.

Air Quality

Franklin County, therefore the ARMD site, is within the South Central Pennsylvania Intrastate Air Quality Control Region AQCR (40 CFR 81.105), which is within the ozone transport region that includes 12 states and Washington, DC. The EPA has designated Franklin County as a maintenance area for the 8-hour ozone National Ambient Air Quality Standard, and in attainment for all other criteria pollutants (40 CFR 81.339).

4.0 DATA QUALITY OBJECTIVES AND CRITERIA

The following section discusses the data quality objectives (DQO) and measurement quality objectives (MQO) identified for this project. LEAD proposes to use the EPA software ProUCL for analysis of variance (ANOVA) consistent with 40 CFR Part 264.97(h) ANOVA analyses. ProUCL provides numerous and varied statistical methods and graphical tools to address environmental sampling and statistical issues. The DQOs and MQOs are consistent with this statistical method.

4.1 DATA QUALITY OBJECTIVES

DQOs are qualitative and quantitative statements developed through the DQO process (EPA, 2006 & EPA, 2000). DQOs clarify the study objective, define the most appropriate data to collect and the conditions under which the data are collected, and specify tolerable limits on decision errors that will be used as the basis for establishing the quantity and quality of data needed to support decision-making. The DQO process is meant to be flexible and iterative. It is not mandatory and is meant as a guide to facilitate a scientific and resource-effective design for acquisition of environmental data. The steps of the DQO process for this project are summarized in Table 1.

4.2 MEASUREMENT QUALITY OBJECTIVES

Analytical results will be evaluated in accordance with precision, accuracy, representativeness, completeness, and comparability (PARCC) parameters to document the quality of the data and to ensure that the data are of sufficient quality to meet the project objectives. Of these PARCC parameters, precision and accuracy will be evaluated quantitatively by collecting the QA/QC samples discussed in Section 7, regarding QA/QC. Each of the PARCC parameters, and how they will be assessed within this project, is also discussed in Section 7.

TABLE 1
DATA QUALITY OBJECTIVES

STEP 1: State the Objective
Baseline and annual data collection basis to evaluate the presence or persistence of contaminant indicator compounds upstream and downstream of the ARMD property, and in soils adjacent to preparation, segmentation and confined burn operations. Continued annual monitoring of select locations allows the compilation of data that can be used for detection monitoring, and future compliance/corrective action analysis, if it should become necessary.
STEP 2: Identify the Decisions
Determine whether significant amount of contaminant indicators are present in existing soil, surface water, groundwater, and sediment/surface runoff within the ARMD property. Determine if change in concentration between baseline and annual sample results exists and if it is of statistical significance.
STEP 3: Identify Inputs to the Decisions
Field data and laboratory analytical results will be collected during up to four baseline sampling events for baseline soil, surface water, groundwater, and sediment/surface runoff; and again during annual sampling events for the locations identified in Figure 3 and Table 2.
STEP 4: Define Study Boundaries
• The study area is defined by the set of sampling locations identified in Figure 3. • Baseline sampling will include a minimum of two sampling events to provide sufficient data for statistical analysis. • Annual sampling will extend through the period identified by PADEP.
STEP 5: Develop Decision Rules
If baseline analyte concentrations are observed prior to startup of the ARMD, further investigation outside this project will be necessary to determine the potential cause(s) of pre-existing elevated concentrations. If change in baseline and annual sample concentrations increases and is statistically significant, and if this condition can be statistically assigned to contributions from within the ARMD, then additional investigation and compliance/corrective action will be determined. If the condition cannot be assigned to the ARMD, no further action is required, except to continue detection monitoring with annual sampling events.

TABLE 1
DATA QUALITY OBJECTIVES

STEP 6: Specify Tolerable Limits on Decision Errors
• Site-specific sampling parameters are predetermined based on historic data collection for open burn events, professional judgment in selecting sampling locations and analytes, and according to the direction of PADEP, for this investigation. Sampling locations have been determined by PADEP with principal decision maker concurrence, based on primary ARMD operations, surface drainage characteristics and nearby surface water locations. Tolerable limits on decision errors cannot be precisely defined at this time. • Potential errors will be reduced by using established SOPs during sampling and analysis; and by closely observing QA/QC protocols for sample collection, handling, and analysis.
STEP 7: Optimize the Sampling Design
Following a comprehensive review of the project baseline and follow up annual results, excluding and/or including specific sampling locations or sampling parameters in future monitoring events will be evaluated.

Notes:

QA/QC Quality assurance and quality control
SOP Standard operating procedure

5.0 MONITORING OF THE ARMD TTC SUMP AND WITNESS ZONE

The Thermal Treatment Chamber (TTC) sump will be equipped with a 6” pipe to allow for visual inspection of the space between the steel liner and the concrete vault for detecting leaks in the steel liner, this inspection space is defined as “the Witness Zone.” The concrete vault acts as secondary containment to the steel liner thereby insuring no liquids contact the soil. Inspection of the witness zone will be performed on a weekly basis. If the ARMD experiences prolonged periods of inactivity, LEAD may refrain from weekly inspections until such time as the ARMD begins operating again. Relief from the weekly inspection requirement must be approved by PADEP prior to suspending any inspections. If liquid is detected in the Witness Zone, a sample will be taken and analyzed for pH using litmus paper or equivalent. If the liquid in the Witness Zone has a pH greater than or equal to 4, it can be assumed that it did not come from the sump and no additional characterization is required. The Witness Zone must be pumped out before resuming operations. If the Witness Zone has a pH less than 4, it must be characterized for appropriate disposal; the sump must be emptied and inspected; and the sump must be repaired for leaks prior to resumption of operations.

The effluent in the sump is processed at the Effluent Handling System onsite. Treated liquid in the TTC waste filtrate tanks is sampled for waste disposal.

6.0 INVESTIGATION OF SOIL, SEDIMENT, SURFACE WATER AND GROUNDWATER

The objective for soil, sediment, surface water, and groundwater sampling is to develop an understanding of existing concentrations of the indicator compounds at the ARMD prior to and following commencement of operations. For the purpose of this plan, reference to surface water and surface water sampling locations for streams are identified and referred to as “surface water.” Stormwater runoff and sediment in areas of stormwater runoff, and their sample locations, are identified and referred to as “stormwater runoff” and “runoff sediment.” This nomenclature is used to avoid confusion between the sample locations and type.

6.1 APPROACH

The ARMD is a contained burn treatment facility. The ARMD is a fully contained unit for demilitarization of rocket motors, the byproduct of which is a non-hazardous solid waste, with no planned releases to surface water, groundwater, soils, stormwater runoff or runoff sediment. However, 40 CFR Part 264 (Standards for Owners and Operators of Hazardous Waste Treatment, Storage, And Disposal Facilities), Subpart F (Releases from Solid Waste Management Units) requires monitoring and response programs.

The monitoring program approach is to establish baseline concentrations of indicator compounds that could result from unplanned releases to soil, surface water, groundwater, stormwater runoff, or runoff sediment, from the rocket motor preparation, segmentation and destruction facilities. Once baseline conditions are established, the program will transition to annual sampling. A reportable indicator compound concentration increase or change from baseline concentrations or from downgradient versus upgradient concentrations in soil, surface water, groundwater, stormwater runoff, or runoff sediment may indicate a release has occurred at the ARMD.

Analytical parameters were selected at the request of PADEP. The initial analytical parameter list was determined based on the system design information; controls; rocket motor combustion byproducts; proximal facility operations; the characteristics of liquid and solid waste on the site; materials used; material storage and handling methods; and PADEP requested compounds. This list may be modified in the future based on investigation results or new waste stream information becoming available or updated.

Baseline levels of potential contaminants will be determined prior to operation of the ARMD. Continued annual sampling will occur following startup of the ARMD. The following list of compounds will be sampled, and was approved by PADEP on March 12, 2016, with the understanding the list is dynamic and could be updated for future sampling events.

- Acidity
- Aluminum (Al)
- Arsenic (As)

- Barium (Ba)
- Bismuth (Bi)
- Cadmium (Cd)
- Chloride (Cl)
- Chromium (Cr)
- Copper (Cu)
- Fluoride (F)
- Iron (Fe)
- Lead (Pb)
- Mercury (Hg)
- Molybdenum (Mo)
- Nitrates/Nitrites (NO₂/NO₃)
- Perchlorate (ClO₄)
- pH (in-situ for water)
- Phosphorus (P)
- Potassium (K)
- Selenium (Se)
- Silica (aka silicon dioxide) (SiO₂)
- Silver (Ag)
- Sulfur/sulfate (S/SO₄)
- Turbidity (in-situ for water)
- Tin (Sn)
- Volatile organic compounds (VOC) – see Appendix A for compound list
- Zirconium (Zr)

The sample locations for this sampling investigation were determined by PADEP after discussion and concurrence with the principal decision makers. A final well location map and well installation details were submitted in the Weston Solutions, Inc. (Weston), Final Work Plan letter report submitted December 2, 2015 and approved by PADEP March 1, 2016. The PADEP approval letter and well installation plan are included as Attachment A. Concurrence with these locations was also confirmed with PADEP in February 2016. The sample location map is provided as Figure 3.

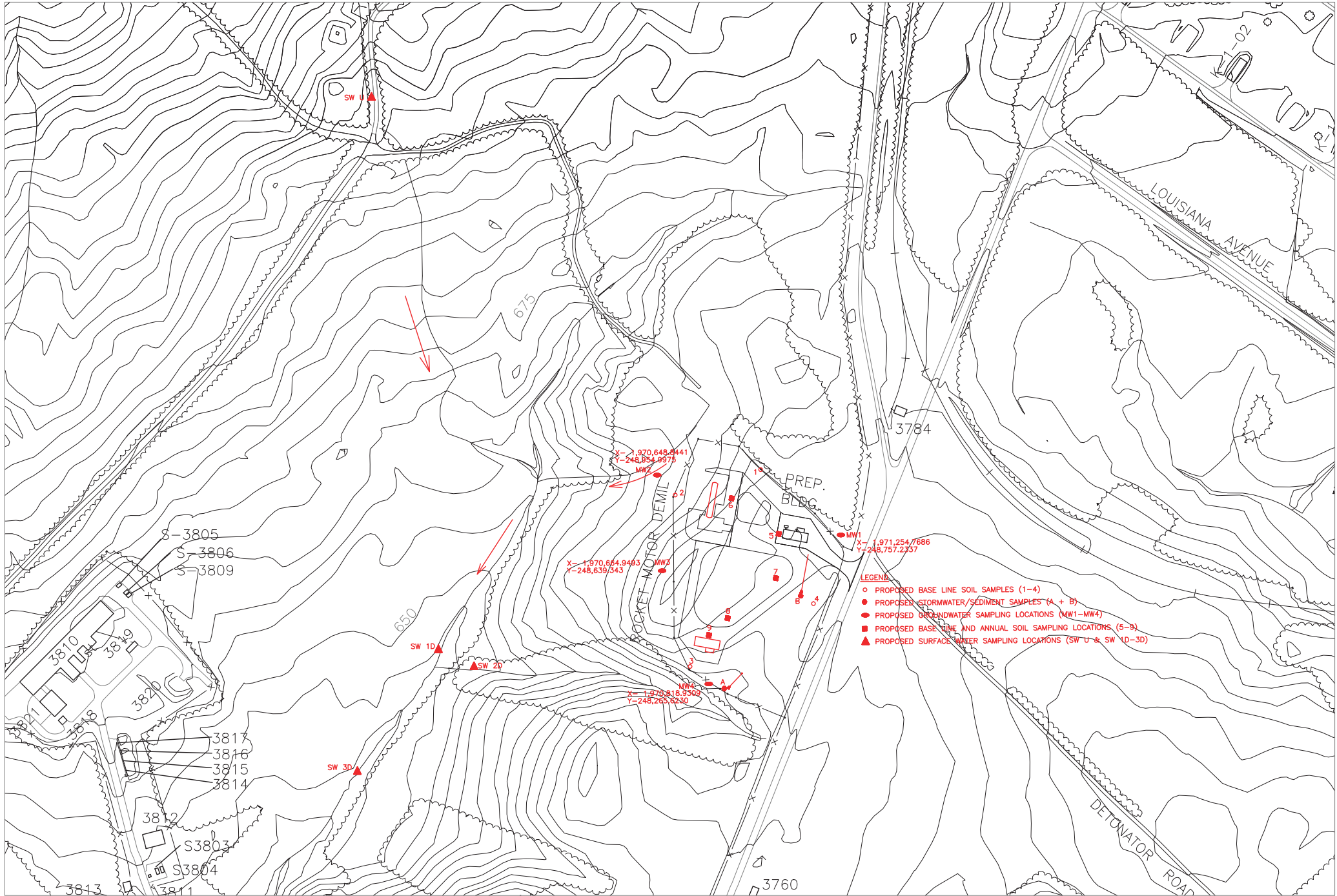


FIGURE 3

APPROVED-				
DATE	10MAR15	REDUCTION OF SOIL SAMPLES		
APPROVED-FIRE CHIEF	REV	DATE	DESCRIPTION	BY
DATE	PROJECT NUMBER		DEPARTMENT OF THE ARMY	
APPROVED-SAFETY OFFICE	DATE:	OFFICE OF ENGINEERING AND PLANNING DIVISION		
DATE	DRAWN BY	WCK	LETTERKENNY ARMY DEPOT	
APPROVED-ENV COORDINATOR	TRACED BY	CHAMBERSBURG, PA		
DATE	CHECKED BY	ARMED FACILITY ENVIRONMENTAL SAMPLING LOCATIONS		
APPROVED-AT/FP OFFICER	SUBMITTED BY	SITE PLAN		
DATE	APPR:	APPROVED	DATE	
APPROVED-CH CONST & TRANS DIV	DATE	CH ENGR & PLAN DIV		
DATE	APPROVED	SCALE: AS SHOWN DATE: 27 JAN 15		
APPROVED-DIR RISK MANAGEMENT	DATE	COMMANDING	DRAWING NUMBER	
DATE	DATE	SK-2073	SHEET 1 OF 1	

Soil baseline samples will be collected at 9 locations along the ARMD fenceline boundary, the three buildings (confined burn building, the preparation building, and the segmentation building), and within the central area of ARMD, to represent background conditions. Annual soil samples will be collected at the same three locations near the buildings used in the ARMD process, and the same two locations of the central area of ARMD, for a total of 5 annual sample locations.

Four monitoring wells will be installed for baseline and annual groundwater sampling, one upgradient and three downgradient of the ARMD. The monitoring wells will be 2 inch diameter, stickup type wells (as opposed to flush mounts at ground surface). Two stormwater runoff locations will be sampled, for stormwater runoff and runoff sediment, to establish baseline and annual stormwater runoff conditions. Four surface water locations identified in the vicinity of the ARMD will be sampled for the baseline and annually. These four surface water locations align with seasonal streams.

Per PADEP request, up to four baseline sampling events will be performed, preferably one in each quarter, prior to start of the ARMD. If quarterly sampling is not feasible with the schedule for construction and start of operation, LEAD will perform up to four baseline sampling events in the time allowed with as much seasonal spacing as is practical. A minimum of two baseline sampling events is needed to provide sufficient data for statistical analysis. Table 2 summarizes the sample location, media type, sampling event and sampling rationale.

TABLE 2
PROPOSED SOIL, SURFACE WATER, GROUNDWATER, STORMWATER RUNOFF, AND
RUNOFF SEDIMENT SAMPLE LOCATION AND SAMPLING RATIONALE

LOCATION NAME (MAP ID)	GENERAL LOCATION	MEDIA	SAMPLING EVENT	RATIONALE
ARMD MW1 (MW1)	EAST OF ARMD	GROUNDWATER	BASELINE ANNUAL	ESTABLISHES UPGRADIENT BASELINE AND ANNUAL CONTAMINANT CONCENTRATIONS
ARMD MW2 (MW2)	WEST OF ARMD	GROUNDWATER	BASELINE ANNUAL	ESTABLISHES DOWNGRADIENT BASELINE AND ANNUAL CONTAMINANT CONCENTRATIONS
ARMD MW3 (MW3)	SOUTHWEST OF ARMD	GROUNDWATER	BASELINE ANNUAL	ESTABLISHES DOWNGRADIENT BASELINE AND ANNUAL CONTAMINANT CONCENTRATIONS
ARMD MW4 (MW4)	SOUTH OF ARMD	GROUNDWATER	BASELINE ANNUAL	ESTABLISHES DOWNGRADIENT BASELINE AND ANNUAL CONTAMINANT CONCENTRATIONS
ARMD SW1 (SW 1D)	SOUTHWEST OF ARMD	SURFACE WATER	BASELINE ANNUAL	ESTABLISHES DOWNGRADIENT BASELINE AND ANNUAL CONTAMINANT CONCENTRATIONS
ARMD SW2 (SW 2D)	SOUTHWEST OF ARMD	SURFACE WATER	BASELINE ANNUAL	ESTABLISHES DOWNGRADIENT BASELINE AND ANNUAL CONTAMINANT CONCENTRATIONS
ARMD SW3 (SW 3D)	SOUTHWEST OF ARMD	SURFACE WATER	BASELINE ANNUAL	ESTABLISHES DOWNGRADIENT BASELINE AND ANNUAL CONTAMINANT CONCENTRATIONS
ARMD SW4 (SW U)	NORTHWEST OF ARMD	SURFACE WATER	BASELINE ANNUAL	ESTABLISHES UPGRADIENT BASELINE AND ANNUAL CONTAMINANT CONCENTRATIONS
ARMD A (A)	SOUTH OF ARMD	STORMWATER RUNOFF	BASELINE ANNUAL	ESTABLISHES BASELINE BACKGROUND AND ANNUAL CONTAMINANT CONCENTRATIONS
		RUNOFF SEDIMENT	BASELINE ANNUAL	ESTABLISHES BASELINE BACKGROUND AND ANNUAL CONTAMINANT CONCENTRATIONS
ARMD B (B)	SOUTHEAST OF ARMD	STORMWATER RUNOFF	BASELINE ANNUAL	ESTABLISHES BASELINE BACKGROUND AND ANNUAL CONTAMINANT CONCENTRATIONS
		RUNOFF SEDIMENT	BASELINE ANNUAL	ESTABLISHES BASELINE BACKGROUND AND ANNUAL CONTAMINANT CONCENTRATIONS
ARMD SL01 (1)	NORTHEAST FENCELINE	SOIL	BASELINE	ESTABLISHES BACKGROUND BASELINE CONTAMINANT CONCENTRATIONS
ARMD SL02 (2)	NORTHWEST FENCELINE	SOIL	BASELINE	ESTABLISHES BACKGROUND BASELINE CONTAMINANT CONCENTRATIONS
ARMD SL03 (3)	SOUTHWEST FENCELINE	SOIL	BASELINE	ESTABLISHES BACKGROUND BASELINE CONTAMINANT CONCENTRATIONS
ARMD SL04 (4)	SOUTHEAST FENCELINE	SOIL	BASELINE	ESTABLISHES BACKGROUND BASELINE CONTAMINANT CONCENTRATIONS

TABLE 2
PROPOSED SOIL, SURFACE WATER, GROUNDWATER, STORMWATER RUNOFF, AND
RUNOFF SEDIMENT SAMPLE LOCATION AND SAMPLING RATIONALE

ARMD SL05 (5)	WEST PREPARATION BLDG	SOIL	BASELINE ANNUAL	ESTABLISHES BACKGROUND BASELINE AND CHANGE IN ANNUAL CONTAMINANT CONCENTRATIONS
ARMD SL06 (6)	EAST BURN CHAMBER BLDG	SOIL	BASELINE ANNUAL	ESTABLISHES BACKGROUND BASELINE AND CHANGE IN ANNUAL CONTAMINANT CONCENTRATIONS
ARMD SL07 (7)	CENTRAL ARMD EAST	SOIL	BASELINE ANNUAL	ESTABLISHES BACKGROUND BASELINE AND CHANGE IN ANNUAL CONTAMINANT CONCENTRATIONS
ARMD SL08 (8)	CENTRAL ARMD SOUTH	SOIL	BASELINE ANNUAL	ESTABLISHES BACKGROUND BASELINE AND CHANGE IN ANNUAL CONTAMINANT CONCENTRATIONS
ARMD SL09 (9)	NORTH SEGMENTATION BLDG	SOIL	BASELINE ANNUAL	ESTABLISHES BACKGROUND BASELINE AND CHANGE IN ANNUAL CONTAMINANT CONCENTRATIONS

6.2 TECHNICAL OR REGULATORY STANDARDS

Existing environmental data for the area of the ARMD is limited. This investigation includes baseline sampling, which will establish existing concentrations of the indicator compounds. At this time operations have not begun, no releases have occurred, and remediation is not applicable. Data from the baseline and annual sampling events will be statistically evaluated together to form the context of indicator compound concentrations and how they exist and change in the environment at the ARMD.

40 CFR Part 264, Subpart F, and by reference Part 261, establishes requirements for Releases from Solid Waste Management Units. Under 264 Subpart F, LEAD will initially implement a detection monitoring program. This ESP, addressing baseline and annual sampling, fulfills the monitoring requirement. If hazardous constituents are detected additional programs or actions may be instituted. For example, per Section 91, if a hazardous constituent is detected (defined as statistically significant evidence of contamination) at a compliance point a compliance monitoring program will be instituted. If a groundwater protection standard is exceeded (defined as statistically significant evidence of increased contamination), a corrective action program will be instituted. If a hazardous constituent exceeds concentration limits in groundwater between the point of compliance and the downgradient facility property boundary, a corrective action program will be instituted.

Section 95 defines the point of compliance as the “vertical surface located at the hydraulically downgradient limit of the waste management area that extends down into the uppermost aquifer underlying the regulated units.” It is the monitoring location at which the groundwater protection standard applies. Section 95 also defines the waste management area. The waste management area limit of the ARMD has been established as the ARMD fenceline. The ARMD points of compliance will be the downgradient wells near the fenceline depicted on Figure 3 (MW2, MW3, or MW4). Actual groundwater flow, established through groundwater elevation measurements (see Section 6.6), will be used to generate contour maps and determine the downgradient flow. Sections 92, 93, and 94 address hazardous constituents and groundwater protection standards. Table 3 presents the maximum constituent concentration for groundwater protection, as established in Subpart F, Section 94.

In determining whether statistically significant evidence of contamination exists, or that there is evidence of a statistically significant increase in contamination, data collected at the compliance points will be compared to background groundwater quality data. Data collected from any of the four wells prior to startup of the ARMD will be treated as background data. Once the ARMD is operating, background data is expected only from the upgradient well (MW1), though this would not preclude the use of data from the downgradient wells.

TABLE 3
40 CFR PART 264.94
MAXIMUM CONCENTRATION OF CONSTITUENTS FOR GROUNDWATER PROTECTION

Constituent	Maximum concentration (mg/L)
Arsenic	0.05
Barium	1.0
Cadmium	0.01
Chromium	0.05
Lead	0.05
Mercury	0.002
Selenium	0.01
Silver	0.05
Endrin (1,2,3,4,10,10-hexachloro-1,7-epoxy 1,4,4a,5,6,7,8,9a-octahydro-1, 4-endo, endo-5,8-dimethano naphthalene)	0.0002
Lindane (1,2,3,4,5,6-hexachlorocyclohexane, gamma isomer)	0.004
Methoxychlor (1,1,1-Trichloro-2,2-bis (p-methoxyphenylethane)	0.1
Toxaphene (C ₁₀ H ₁₀ Cl ₆ , Technical chlorinated camphene, 67-69 percent chlorine)	0.005
2,4-D (2,4-Dichlorophenoxyacetic acid)	0.1
2,4,5-TP Silvex (2,4,5-Trichlorophenoxypropionic acid)	0.01

Source: (CFR 2016a)

While data will generally be reviewed relative to background (baseline) versus annual and/or upgradient versus downgradient, the data will also be compared to Pennsylvania and EPA requirements and guidance for groundwater, surface water, and soil quality. This comparison is for reference only and will be performed to establish existing baseline concentrations do not show elevated concentrations; or that if elevated concentrations exist in the background that they cannot be attributed to the ARMD. It should be noted that samples containing high concentrations will be resampled within 10 days to confirm the analysis, or identify the sample as a potential outlier. This comparison may spot potential increasing concentration trends, and thus minimize environmental impacts by taking action before significant degradation occurs and remediation would be necessary. The applicability of Pennsylvania and EPA standards and guidelines to the ARMD can be established by PADEP at a later date should it be deemed necessary.

Table 4 presents Pennsylvania and EPA standards and guidelines for groundwater, surface water, and soil quality. Pennsylvania sets Medium-Specific Concentrations (MSC) for soil and water. EPA sets federal maximum contaminant levels (MCL) or secondary standards, which are set national primary drinking water standards, and secondary drinking water standards, and industrial soil screening levels (SSL). Data collected for this project will be of a resolution sufficient for comparison to the values presented in Table 4. This is further discussed in Section 6.4 Analytical Methods and Detection Limits.

TABLE 4
PENNSYLVANIA AND U.S. ENVIRONMENTAL PROTECTION AGENCY
STANDARDS AND GUIDELINES

Pennsylvania Code 250, Appendix A, Medium-Specific Concentrations					
Analyte	CASRN	Groundwater MSC		Soil MSC	
		Used NR	Units	NR	Units
Aluminum	7429-90-5	200	µg/L	190,000 C	mg/kg
Arsenic	7440-38-2	10	µg/L	53 G 190,000 C	mg/kg mg/kg
0-2 feet					
2-15 feet					
Barium	7440-39-3	2,000	µg/L	190,000 C	mg/kg
Bismuth	NA	NA	NA	NA	NA
Cadmium	7440-43-9	5	µg/L	1,400 G 190,000 C	mg/kg mg/kg
0-2 feet					
2-15 feet					
Chloride	7647-14-5	250,000	µg/L	NA	NA
Chromium (III)	16065-83-1	NA	NA	190,000 C	mg/kg
Chromium (VI)	18540-29-9	NA	NA	8,400 G 20,000 N	mg/kg mg/kg
0-2 feet					
2-15 feet					
Chromium (Total)	7440-47-3	100	µg/L	NA	NA
Copper	7440-50-8	1,000	µg/L	100,000 G 190,000 C	mg/kg mg/kg
0-2 feet					
2-15 feet					
Fluoride	16984-48-8	4,000	µg/L	110,000 G 190,000 G	mg/kg mg/kg
0-2 feet					
2-15 feet					
Iron	7439-89-6	300	µg/L	190,000 C	mg/kg
Lead	7439-92-1	5	µg/L	1,000 S 190,000 C	mg/kg mg/kg
0-2 feet					
2-15 feet					
Mercury	7439-97-6	2	µg/L	450 G 190,000 C	mg/kg mg/kg
0-2 feet					
2-15 feet					
Molybdenum	7439-98-7	40	µg/L	14,000 G 190,000 C	mg/kg mg/kg
0-2 feet					
2-15 feet					
Nitrate as nitrogen	14797-55-8	10,000	µg/L	NA	NA
Nitrite as nitrogen	14797-65-0	1,000	µg/L	NA	NA
Perchlorate	7790-98-9	15	µg/L	2,000 G 190,000 C	mg/kg mg/kg
0-2 feet					
2-15 feet					
Phosphorus	NA	NA	NA	NA	NA
Potassium	NA	NA	NA	NA	NA
Selenium	7782-49-2	50	µg/L	14,000 G 190,000 C	mg/kg mg/kg
0-2 feet					
2-15 feet					
Silica	NA	NA	NA	NA	NA
Silver	7440-22-4	100	µg/L	14,000 G 190,000 C	mg/kg mg/kg
0-2 feet					
2-15 feet					

TABLE 4 (Continued)
PENNSYLVANIA AND U.S. ENVIRONMENTAL PROTECTION AGENCY
STANDARDS AND GUIDELINES

Pennsylvania Code 250, Appendix A, Medium-Specific Concentrations					
Analyte	CASRN	Groundwater MSC		Soil MSC	
		Used NR	Units	NR	Units
Sulfur/sulfate	7757-82-6	250,000	µg/L	NA	NA
Tin	7440-31-5	61,000	µg/L	190,000 C	mg/kg
Zirconium	NA	NA	NA	NA	NA
Notes: Assume groundwater TDS is less than or equal to 2,500 Aluminum, chloride, iron, and sulfur/sulfate groundwater MSC is secondary contaminant designated as a secondary maximum contaminant level. C Cap G Ingestion N Inhalation S SEGH Model MSC Medium-specific concentration NR Non-residential Used Used aquifer					
Source: (Pa. Code 2016b)					
U.S. Environmental Protection Agency Standards and Screening Levels					
Analyte	Water			Soil	
	MCLG	MCL	Secondary Standard	Soil Screening Level (mg/kg)	
Aluminum	NA	NA	0.05 – 0.2 mg/L	1.1E+06	
Arsenic	0 mg/L	0.010 mg/L	NA	3.0E+00	
Barium	2 mg/L	2 mg/L	NA	2.2E+05	
Bismuth	NA	NA	NA	NA	
Cadmium	0.005 mg/L	0.005 mg/L	NA	9.8E+02	
Chloride	NA	NA	250 mg/L	NA	
Chromium (III)	NA	NA	NA	1.8E+06	
Chromium (VI)	NA	NA	NA	6.3E+00	
Chromium (Total)	0.1 mg/L	0.1 mg/L	NA	NA	
Copper	1.3 mg/L	1.3 mg/L Action Level	1.0 mg/L	4.7E+04	
Fluoride	4.0 mg/L	4.0 mg/L	2.0 mg/L	4.7E+04	
Iron	NA	NA	0.3 mg/L	8.2E+05	
Lead and Compounds	0 mg/L	0.015 mg/L Action Level	NA	8.0E+02	
Mercury (elemental)	0.002 mg/L	0.002 mg/L	NA	4.6E+01	
Molybdenum	NA	NA	NA	5.8E+03	
Nitrate (measured as nitrogen)	10 mg/L	10 mg/L	NA	1.9E+05	
Nitrite (measured as nitrogen)	1 mg/L	1 mg/L	NA	1.2E+04	
Perchlorate	15 µg/L	Pending	NA	8.2E+01	
pH	NA	NA	6.5 – 8.5	NA	
Phosphorus (White)	NA	NA	NA	2.3E+01	
Potassium	NA	NA	NA	NA	
Selenium	0.05 mg/L	0.05 mg/L	NA	5.8E+03	
Silica	NA	NA	NA	1.8E+07	
Silver	NA	NA	0.1 mg/L	5.8E+03	

TABLE 4 (Continued)
PENNSYLVANIA AND U.S. ENVIRONMENTAL PROTECTION AGENCY
STANDARDS AND GUIDELINES

U.S. Environmental Protection Agency Standards and Screening Levels				
Analyte	Water			Soil
	MCLG	MCL	Secondary Standard	Soil Screening Level (mg/kg)
Sulfur/sulfate	NA	NA	250 mg/L	NA
Tin	NA	NA	NA	7.0E+05
Zirconium	NA	NA	NA	9.3E+01
Notes: MCL Maximum Contaminant Level – The highest level of a contaminant that is allowed in drinking water. MCLs are set as close to MCLGs as feasible using the best available treatment technology and taking cost into consideration. MCLs are enforceable standards. MCLG Maximum Contaminant Level Goal – The level of a contaminant in drinking water below which there is no known or expected risk to health. MCLGs allow for a margin of safety and are non-enforceable public health goals. mg/L milligrams per liter Secondary Standard – National Secondary Drinking Water Regulations are non-enforceable guidelines regulating contaminants that may cause cosmetic effects or aesthetic effects in drinking water. Soil screening level is for industrial soil, and is taken from Region 3 Regional Screening Levels table for contaminants at CERCLA sites. Sources: (EPA 2016) and (EPA 2015)				

6.3 PROJECT TIMELINE

Table 5 summarizes the outline for schedule of sampling, analysis, and reporting for this project. Dates and times for these milestones will be determined by the principal decision makers at a future date. Monitoring well installation is not addressed in this ESP, but is required prior to sampling activities. Monitoring well installation will be performed by a licensed driller according to PADEP codes and regulations and will generally follow Chapters 2 and 3 of the PADEP “Groundwater Guidance Manual” (PADEP, 2001). The well installation plan including groundwater monitoring wells relative to this project was submitted to PADEP in December 2015 and provides additional details, see Attachment A (Weston, 2015).

TABLE 5
IMPLEMENTATION SCHEDULE FOR SAMPLING, ANALYSIS, AND REPORTING

Milestone	Due Date	Anticipated Date
Monitoring Well Installation	Following well installation plan approval by PADEP	May 16, 2016
Field investigation	TBD days following monitoring well installation	TBD
Baseline summary report	60 days after completion of baseline field work	TBD
Electronic data deliverables	TBD days after completion of baseline field work	TBD
Annual Sampling & Deliverables	As required by PADEP	TBD

6.4 ANALYTICAL METHODS AND REPORTING LIMITS

Pennsylvania Code presents information on establishing analytical methods, detection limits, and medium-specific concentrations under Chapters 16 and 250. The information in these codes has been referenced to establish the analytical methods and practical quantitation limits (PQL) used for this project.

Per Pennsylvania Code, the EPA estimated quantitation limits (EQL) found in the most current version of the EPA RCRA Manual SW-846 for soil listed as “low level soil” and for groundwater listed as “groundwater,” will be selected as the project PQLs.

Specifically the PQLs for inorganic compounds will be those values listed for analysis by inductively coupled plasma (ICP). However, lead, cadmium, and arsenic (all on the sample compound list) will use the EQL values listed for the atomic absorption graphite furnace methods for water as the project PQL. The mercury PQL will use the EQL value listed for the cold vapor method. For organic compounds, the PQLs will be the EQLs listed for the gas chromatography/mass spectrometry.

Pennsylvania Code presents approved EPA analytical methods and detection limits in Title 25, Chapter 16, Subchapter B. Additionally, analytical methods and their associated reporting limits were presented in the depot-wide QAPP that supports environmental analytical services for environmental activities at LEAD (Weston, 2003), and which have been included in Attachment B.

The following parameters will be sampled for in soil, surface water, groundwater, stormwater runoff and runoff sediment. Anticipated analyses for these samples include the methods listed. Additionally, water pH will be measured in-situ in the flow through cell for groundwater or from the surface water body. Dissolved metals, including bismuth and zirconium, will be field filtered with a disposable, inline, 0.45µm filter at the time of sample collection. Turbidity will also be measured in-situ.

- Acidity – SW846 Method 9045D/2310B or EPA 3050B/3052
- Aluminum (Al) – SW846 Method 6020B or EPA 200.7/200.8
- Arsenic (As) – SW846 Method 7010 or EPA 200.9
- Barium (Ba) – SW846 Method 6020B or EPA 200.7/200.8
- Bismuth (Bi) – SW846 Method 6010 or EPA 200.7
- Cadmium (Cd) – SW846 Method 7010 or EPA 200.9
- Chloride (Cl) – SW846 Method 9056A or EPA 300.0/300.1/300.6
- Chromium (Cr) – SW846 Method 6020B or EPA 200.7/200.8
- Copper (Cu) – SW846 Method 6020B or EPA 200.7/200.8
- Fluoride (F) – SW846 Method 9056A or EPA 300.0/300.1/300.6
- Iron (Fe) – SW846 Method 6020B or EPA 200.7
- Lead (Pb) – SW846 Method 7010 or EPA 200.9
- Mercury (Hg) – SW846 Method 7470A liquid & 7471B solids or EPA 245.1/245.2
- Molybdenum (Mo) – SW846 Method 6020B or EPA 200.7/200.8

- Nitrates/Nitrites (NO_2/NO_3) – SW846 Method 9056A or EPA 300.0/300.1/300.6
- Perchlorate (ClO_4) – SW846 Method 6850/6860
- Phosphorus (P) – SW846 Method 6010 or EPA 300.0
- Potassium (K) – SW846 Method 6020B or EPA 300.7
- Selenium (Se) – SW846 Method 6020B or EPA 200.7/200.8
- Silica (aka silicon dioxide) (SiO_2) – 6010D
- Silver (Ag) – SW846 Method 6020B or EPA 200.7/200.8
- Sulfur/sulfate (S/SO_4) – SW846 Method 9056A or EPA 300.0
- Tin (Sn) – SW846 Method 6020B or EPA 200.7
- Volatile organic compounds (VOC) – SW846 Method 8260C
- Zirconium (Zr) - SW846 Method 6200

Potential laboratories for this investigation will be selected from a list of PADEP accredited laboratories developed by CONTRACTOR. LEAD will select and approve the final laboratory or laboratories for this field investigation. Table 6 presents the anticipated sample volume, containers, hold times and preservatives for each sample media. The CONTRACTOR and contracted laboratory will be responsible for finalizing the analytical method, containers, holding times, and reporting limits.

TABLE 6
SAMPLES VOLUMES, CONTAINERS, PRESERVATIVES, AND HOLDING TIMES

Analysis	Sample Volume	Container	Preservative	Holding Time
Soil/Sediment				
Metals	4 oz	Glass jar w/ Teflon lid	NA, Cool to 4 °C	6 months; Hg 28 days
VOC	4 oz 3 – Encore Sampler	Glass jar w/ Teflon lid OR Encore sample	NA, Cool to 4 °C No headspace	14 days Encore – 48 hr to extraction
Perchlorate	4 oz	Glass jar w/ Teflon lid	NA, Cool to 4 °C	28 days
Acidity	4 oz	Glass jar OR quart bag	NA, Cool to 4 °C	14 days
Bismuth	4 oz	Plastic jar OR quart bag	NA, Cool to 4 °C	6 months
Phosphorus	4 oz	Glass jar w/ Teflon lid	NA, Cool to 4 °C	28 days
Nitrate/Nitrite	4 oz	Glass jar w/ Teflon lid	NA, Cool to 4 °C	7 days
Sulfur/sulfate	4 oz	Glass jar w/ Teflon lid	NA, Cool to 4 °C	28 days
Chloride	4 oz	Glass jar w/ Teflon lid	NA, Cool to 4 °C	28 days
Fluoride	4 oz	Glass jar w/ Teflon lid	NA, Cool to 4 °C	28 days
Silica (Silicon dioxide)	4 oz	Glass jar w/ Teflon lid	NA, Cool to 4 °C	6 months
Zirconium	4 oz	Glass jar w/ Teflon lid	NA, Cool to 4 °C	6 months
Water				
Metals	1 L	Plastic bottle	HNO ₃ , Cool to 4 °C	6 months; Hg 28 days
VOC	3 – 40 mL VOA	Amber, septum-capped glass VOA vial	HCL, Cool to 4 °C No headspace	14 days
Perchlorate	1 L	Glass / plastic bottle	NA, Cool to 4 °C	28 days
Acidity	250 mL	Glass / plastic bottle	NA, Cool to 4 °C	14 days
Bismuth	500 mL	Plastic bottle	HNO ₃ , Cool to 4 °C	6 months
Phosphorus	250 mL	Glass / plastic bottle	H ₂ SO ₄ , Cool to 4 °C	28 days
Nitrate/Nitrite	125 mL	Glass / plastic bottle	H ₂ SO ₄ , Cool to 4 °C	28 days
Sulfur/sulfate	125 mL	Glass / plastic bottle	NA, Cool to 4 °C	28 days
Chloride	125 mL	Glass / plastic bottle	NA, Cool to 4 °C	28 days
Fluoride	250mL	Glass / plastic bottle	NA, Cool to 4 °C	28 days
Silica (Silicon dioxide)	250 mL	Plastic bottle	HNO ₃ , Cool to 4 °C	6 months
Zirconium	250 mL	Plastic bottle	HNO ₃ , Cool to 4 °C	6 months

6.5 FIELD DOCUMENTATION

Complete and accurate documentation is essential to demonstrate that field measurement and sampling procedures are carried out as described in this ESP. Field personnel will use permanently bound field logbooks with sequentially numbered pages to record and document field activities. The logbook will list the contract name and number, site name, and names of subcontractors, the service client, and the project manager. At a minimum, the following information will be recorded in the field logbook:

- Name and affiliation of all on-site personnel or visitors
- Weather conditions during the field activity
- Summary of daily activities and significant events
- Notes of conversations with coordinating officials
- References to other field logbooks or forms that contain specific information
- Discussions of problems encountered and their resolution
- Discussions of deviations from the ESP or other governing documents
- Description of all photographs taken, if any

The field team will also use various field forms. CONTRACTOR will provide specific forms for this sampling investigation, however anticipated field forms may include:

- Field Instrument Calibration Log
- Chain-of-Custody
- Tailgate Safety Meeting Form
- Daily Quality Control Report
- Audit Report
- Corrective Action Form

6.6 SAMPLE COLLECTION

Soil, surface water, groundwater, stormwater runoff and runoff sediment, will be collected as described in the following section. Samples will be collected from the locations identified in Figure 3 and Table 2, and will be analyzed for the parameters identified in Sections 6.1 and 6.4. Table 7 provides proposed field sample identification and QA/QC sample collection for the baseline and annual sampling events. Sample identification is further addressed in Section 6.8.1. Figure 3 shows local streams in the region surrounding the ARMD as an indicator of surface water and potential groundwater flow (subject to field verification and initial well installation). Table 6 presents the sample compound, sample container, holding times, preservative and handling temperature.

TABLE 7
PROPOSED SOIL, SURFACE WATER, GROUNDWATER, STORMWATER RUNOFF, AND
RUNOFF SEDIMENT SAMPLE FIELD SAMPLING IDENTIFICATION AND COLLECTION

BASELINE SAMPLES FOR SAMPLING EVENT 1 (PRIOR TO STARTUP)				
LOCATION	MEDIA	SAMPLE ID	TYPE	COMMENTS
ARMD MW1	GROUNDWATER	ARMDMW1-DATE	ENVIRONMENTAL	
		ARMDMW1-DATE-D	QA/QC	DUPLICATE
ARMD MW2	GROUNDWATER	ARMDMW2-DATE	ENVIRONMENTAL	
ARMD MW3	GROUNDWATER	ARMDMW3-DATE	ENVIRONMENTAL QA/QC	MS/MSD – COLLECT ADDITIONAL SAMPLE VOLUME
ARMD MW4	GROUNDWATER	ARMDMW4-DATE	ENVIRONMENTAL	
ARMD SW1	SURFACE WATER	ARMDSW1-DATE	ENVIRONMENTAL	
ARMD SW2	SURFACE WATER	ARMDSW2-DATE	ENVIRONMENTAL	
		ARMDSW2-DATE-D	QA/QC	DUPLICATE
ARMD SW3	SURFACE WATER	ARMDSW3-DATE	ENVIRONMENTAL	
ARMD SW4	SURFACE WATER	ARMDSW4-DATE	ENVIRONMENTAL	
ARMD A	STORMWATER RUNOFF	ARMDARW-DATE	ENVIRONMENTAL	
		ARMDARW-DATE-D	QA/QC	DUPLICATE
ARMD B	RUNOFF SEDIMENT	ARMDARS-DATE	ENVIRONMENTAL	
	STORMWATER RUNOFF	ARMDBRW-DATE	ENVIRONMENTAL	
	RUNOFF SEDIMENT	ARMDBRS-DATE	ENVIRONMENTAL	
		ARMDBRS-DATE-D	QA/QC	DUPLICATE
ARMD SL01	SOIL	ARMDSL01-DATE	ENVIRONMENTAL	
ARMD SL02	SOIL	ARMDSL02-DATE	ENVIRONMENTAL	
ARMD SL03	SOIL	ARMDSL03-DATE	ENVIRONMENTAL	
ARMD SL04	SOIL	ARMDSL04-DATE	ENVIRONMENTAL	
	SOIL	ARMDSL04-DATE-D	QA/QC	DUPLICATE
ARMD SL05	SOIL	ARMDSL05-DATE	ENVIRONMENTAL	
ARMD SL06	SOIL	ARMDSL06-DATE	ENVIRONMENTAL	
ARMD SL07	SOIL	ARMDSL07-DATE	ENVIRONMENTAL	
ARMD SL08	SOIL	ARMDSL08-DATE	ENVIRONMENTAL QA/QC	MS/MSD – COLLECT ADDITIONAL SAMPLE VOLUME
ARMD SL09	SOIL	ARMDSL09-DATE	ENVIRONMENTAL	
NA	LABORATORY WATER	ARMDTB-DATE-XX	QA/QC	TRIP BLANK

TABLE 7 (continued)
PROPOSED SOIL, SURFACE WATER, GROUNDWATER, STORMWATER RUNOFF, AND
RUNOFF SEDIMENT SAMPLE FIELD SAMPLING IDENTIFICATION AND COLLECTION

BASELINE SAMPLES FOR SAMPLING EVENT 2 (PRIOR TO STARTUP)				
LOCATION	MEDIA	SAMPLE ID	TYPE	COMMENTS
ARMD MW1	GROUNDWATER	ARMDMW1-DATE	ENVIRONMENTAL	
ARMD MW2	GROUNDWATER	ARMDMW2-DATE	ENVIRONMENTAL	
		ARMDMW2-DATE-D	QA/QC	DUPLICATE
ARMD MW3	GROUNDWATER	ARMDMW3-DATE	ENVIRONMENTAL	
ARMD MW4	GROUNDWATER	ARMDMW4-DATE	ENVIRONMENTAL QA/QC	MS/MSD – COLLECT ADDITIONAL SAMPLE VOLUME
ARMD SW1	SURFACE WATER	ARMDSW1-DATE	ENVIRONMENTAL	
ARMD SW2	SURFACE WATER	ARMDSW2-DATE	ENVIRONMENTAL	
ARMD SW3	SURFACE WATER	ARMDSW3-DATE	ENVIRONMENTAL	
		ARMDSW3-DATE-D	QA/QC	DUPLICATE
ARMD SW4	SURFACE WATER	ARMDSW4-DATE	ENVIRONMENTAL	
ARMD A	STORMWATER RUNOFF	ARMDARW-DATE	ENVIRONMENTAL	
	RUNOFF SEDIMENT	ARMDARS-DATE	ENVIRONMENTAL	
		ARMDARS-DATE-D	QA/QC	DUPLICATE
ARMD B	STORMWATER RUNOFF	ARMDBRW-DATE	ENVIRONMENTAL	
		ARMDBRW-DATE-D	QA/QC	DUPLICATE
	RUNOFF SEDIMENT	ARMDBRS-DATE	ENVIRONMENTAL	
ARMD SL01	SOIL	ARMDSL01-DATE	ENVIRONMENTAL	
ARMD SL02	SOIL	ARMDSL02-DATE	ENVIRONMENTAL	
		ARMDSL02-DATE-D	QA/QC	DUPLICATE
ARMD SL03	SOIL	ARMDSL03-DATE	ENVIRONMENTAL QA/QC	MS/MSD – COLLECT ADDITIONAL SAMPLE VOLUME
ARMD SL04	SOIL	ARMDSL04-DATE	ENVIRONMENTAL	
ARMD SL05	SOIL	ARMDSL05-DATE	ENVIRONMENTAL	
ARMD SL06	SOIL	ARMDSL06-DATE	ENVIRONMENTAL	
ARMD SL07	SOIL	ARMDSL07-DATE	ENVIRONMENTAL	
ARMD SL08	SOIL	ARMDSL08-DATE	ENVIRONMENTAL	
ARMD SL09	SOIL	ARMDSL09-DATE	ENVIRONMENTAL	
NA	LABORATORY WATER	ARMDTB-DATE-XX	QA/QC	TRIP BLANK

TABLE 7 (continued)
PROPOSED SOIL, SURFACE WATER, GROUNDWATER, STORMWATER RUNOFF, AND
RUNOFF SEDIMENT SAMPLE FIELD SAMPLING IDENTIFICATION AND COLLECTION

BASELINE SAMPLES FOR SAMPLING EVENT 3 (PRIOR TO STARTUP) – IF APPLICABLE				
LOCATION	MEDIA	SAMPLE ID	TYPE	COMMENTS
ARMD MW1	GROUNDWATER	ARMDMW1-DATE	ENVIRONMENTAL QA/QC	MS/MSD – COLLECT ADDITIONAL SAMPLE VOLUME
ARMD MW2	GROUNDWATER	ARMDMW2-DATE	ENVIRONMENTAL	
ARMD MW3	GROUNDWATER	ARMDMW3-DATE	ENVIRONMENTAL	
		ARMDMW3-DATE-D	QA/QC	DUPLICATE
ARMD MW4	GROUNDWATER	ARMDMW4-DATE	ENVIRONMENTAL	
ARMD SW1	SURFACE WATER	ARMDSW1-DATE	ENVIRONMENTAL	
ARMD SW2	SURFACE WATER	ARMDSW2-DATE	ENVIRONMENTAL	
ARMD SW3	SURFACE WATER	ARMDSW3-DATE	ENVIRONMENTAL	
ARMD SW4	SURFACE WATER	ARMDSW4-DATE	ENVIRONMENTAL	
		ARMDSW4-DATE-D	QA/QC	DUPLICATE
ARMD A	STORMWATER RUNOFF	ARMDARW-DATE	ENVIRONMENTAL	
	RUNOFF SEDIMENT	ARMDARS-DATE	ENVIRONMENTAL	
		ARMDARS-DATE-D	QA/QC	DUPLICATE
ARMD B	STORMWATER RUNOFF	ARMDBRW-DATE	ENVIRONMENTAL	
		ARMDBRW-DATE-D	QA/QC	DUPLICATE
	RUNOFF SEDIMENT	ARMDBRS-DATE	ENVIRONMENTAL	
ARMD SL01	SOIL	ARMDSL01-DATE	ENVIRONMENTAL	
ARMD SL02	SOIL	ARMDSL02-DATE	ENVIRONMENTAL	
ARMD SL03	SOIL	ARMDSL03-DATE	ENVIRONMENTAL	
ARMD SL04	SOIL	ARMDSL04-DATE	ENVIRONMENTAL	
ARMD SL05	SOIL	ARMDSL05-DATE	ENVIRONMENTAL QA/QC	MS/MSD – COLLECT ADDITIONAL SAMPLE VOLUME
ARMD SL06	SOIL	ARMDSL06-DATE	ENVIRONMENTAL	
ARMD SL07	SOIL	ARMDSL07-DATE	ENVIRONMENTAL	
		ARMDSL07-DATE-D	QA/QC	DUPLICATE
ARMD SL08	SOIL	ARMDSL08-DATE	ENVIRONMENTAL	
ARMD SL09	SOIL	ARMDSL09-DATE	ENVIRONMENTAL	
NA	LABORATORY WATER	ARMDTB-DATE-XX	QA/QC	TRIP BLANK

TABLE 7 (continued)
PROPOSED SOIL, SURFACE WATER, GROUNDWATER, STORMWATER RUNOFF, AND
RUNOFF SEDIMENT SAMPLE FIELD SAMPLING IDENTIFICATION AND COLLECTION

BASELINE SAMPLES FOR SAMPLING EVENT 4 (PRIOR TO STARTUP) – IF APPLICABLE				
LOCATION	MEDIA	SAMPLE ID	TYPE	COMMENTS
ARMD MW1	GROUNDWATER	ARMDMW1-DATE	ENVIRONMENTAL	
ARMD MW2	GROUNDWATER	ARMDMW2-DATE	ENVIRONMENTAL QA/QC	MS/MSD – COLLECT ADDITIONAL SAMPLE VOLUME
ARMD MW3	GROUNDWATER	ARMDMW3-DATE	ENVIRONMENTAL	
ARMD MW4	GROUNDWATER	ARMDMW4-DATE	ENVIRONMENTAL	
		ARMDMW4-DATE-D	QA/QC	DUPLICATE
ARMD SW1	SURFACE WATER	ARMDSW1-DATE	ENVIRONMENTAL	
		ARMDSW1-DATE-D	QA/QC	DUPLICATE
ARMD SW2	SURFACE WATER	ARMDSW2-DATE	ENVIRONMENTAL	
ARMD SW3	SURFACE WATER	ARMDSW3-DATE	ENVIRONMENTAL	
ARMD SW4	SURFACE WATER	ARMDSW4-DATE	ENVIRONMENTAL	
ARMD A	STORMWATER RUNOFF	ARMDARW-DATE	ENVIRONMENTAL	
		ARMDARW-DATE-D	QA/QC	DUPLICATE
	RUNOFF SEDIMENT	ARMDARS-DATE	ENVIRONMENTAL	
ARMD B	STORMWATER RUNOFF	ARMDBRW-DATE	ENVIRONMENTAL	
		ARMDBRS-DATE	ENVIRONMENTAL	
	RUNOFF SEDIMENT	ARMDBRS-DATE-D	QA/QC	DUPLICATE
ARMD SL01	SOIL	ARMDSL01-DATE	ENVIRONMENTAL QA/QC	MS/MSD – COLLECT ADDITIONAL SAMPLE VOLUME
ARMD SL02	SOIL	ARMDSL02-DATE	ENVIRONMENTAL	
ARMD SL03	SOIL	ARMDSL03-DATE	ENVIRONMENTAL	
ARMD SL04	SOIL	ARMDSL04-DATE	ENVIRONMENTAL	
ARMD SL05	SOIL	ARMDSL05-DATE	ENVIRONMENTAL	
ARMD SL06	SOIL	ARMDSL06-DATE	ENVIRONMENTAL	
		ARMDSL06-DATE-D	QA/QC	DUPLICATE
ARMD SL07	SOIL	ARMDSL07-DATE	ENVIRONMENTAL	
ARMD SL08	SOIL	ARMDSL08-DATE	ENVIRONMENTAL	
ARMD SL09	SOIL	ARMDSL09-DATE	ENVIRONMENTAL	
NA	LABORATORY WATER	ARMDTB-DATE-XX	QA/QC	TRIP BLANK

TABLE 7 (continued)
PROPOSED SOIL, SURFACE WATER, GROUNDWATER, STORMWATER RUNOFF, AND
RUNOFF SEDIMENT SAMPLE FIELD SAMPLING IDENTIFICATION AND COLLECTION

ANNUAL SAMPLES FOR SAMPLING EVENT 1 (FOLLOWING OPERATIONS START UP)				
LOCATION	MEDIA	SAMPLE ID	TYPE	COMMENTS
ARMD MW1	GROUNDWATER	ARMDMW1-DATE	ENVIRONMENTAL	
		ARMDMW1-DATE-D	QA/QC	DUPLICATE
ARMD MW2	GROUNDWATER	ARMDMW2-DATE	ENVIRONMENTAL	
ARMD MW3	GROUNDWATER	ARMDMW3-DATE	ENVIRONMENTAL QA/QC	MS/MSD – COLLECT ADDITIONAL SAMPLE VOLUME
ARMD MW4	GROUNDWATER	ARMDMW4-DATE	ENVIRONMENTAL	
ARMD SW1	SURFACE WATER	ARMDSW1-DATE	ENVIRONMENTAL	
ARMD SW2	SURFACE WATER	ARMDSW2-DATE	ENVIRONMENTAL	
		ARMDSW2-DATE-D	QA/QC	DUPLICATE
ARMD SW3	SURFACE WATER	ARMDSW3-DATE	ENVIRONMENTAL	
ARMD SW4	SURFACE WATER	ARMDSW4-DATE	ENVIRONMENTAL	
ARMD A	STORMWATER RUNOFF	ARMDARW-DATE	ENVIRONMENTAL	
		ARMDARW-DATE-D	QA/QC	DUPLICATE
ARMD B	RUNOFF SEDIMENT	ARMDARS-DATE	ENVIRONMENTAL	
	STORMWATER RUNOFF	ARMDBRW-DATE	ENVIRONMENTAL	
		ARMDBRS-DATE	ENVIRONMENTAL	
ARMD SL05	SOIL	ARMDSL05-DATE	ENVIRONMENTAL	
		ARMDSL05-DATE-D	QA/QC	DUPLICATE
ARMD SL06	SOIL	ARMDSL06-DATE	ENVIRONMENTAL	
ARMD SL07	SOIL	ARMDSL07-DATE	ENVIRONMENTAL	
ARMD SL08	SOIL	ARMDSL08-DATE	ENVIRONMENTAL	
ARMD SL09	SOIL	ARMDSL09-DATE	ENVIRONMENTAL QA/QC	MS/MSD – COLLECT ADDITIONAL SAMPLE VOLUME
NA	LABORATORY WATER	ARMDTB-DATE-XX	QA/QC	TRIP BLANK
NOTES: ID Identification MS/MSD Matrix spike./matrix spike duplicate QA/QC Quality assurance/quality control “DATE” in the sample ID is the date the sample is collected and should be represented as an eight digit date beginning with the month, day, and year. For example ARMDMW1-06142017 “XX” in the sample ID is a sequential number, beginning with 01, in the event that more than one QA/QC of this type is collected on the same day. All samples will undergo laboratory analysis for: AL, AG, AS, BA, BI, CD, CL, CR, CU, CLO ₄ , F, FE, HG, K, NO ₂ , NO ₃ , MO, P, PB, PH, S/SO ₄ , SE, SI, SN, VOC, ZR “Annual” sampling events will be performed quarterly following start up of the ARMD for one year. After the first year, sampling events will occur annually. Only sampling information relative to the first Annual Sampling Event is provided here. The CONTRACTOR will determine field identification and collection information for future annual events as appropriate.				

All samples will be collected to avoid cross-contamination and worker exposure using new nitrile gloves for each sample and at each sample location. Additionally, samples will be collected in new, clean, laboratory-supplied sample containers. Disposable or dedicated equipment will be used whenever possible. Upon collection, all sample containers will be labeled, placed in protective material, and cooled and maintained at 4 °C.

6.6.1 Soil Samples

Baseline soil sampling will occur at 9 locations, as identified in Figure 3. Up to four baseline sampling events will be performed prior to startup of the ARMD. A total of 9 soil samples, plus additional QA/QC samples, will be collected from these locations to establish baseline conditions. For subsequent annual sampling, after the ARMD operations have begun, soil samples will be collected from 5 annual sampling locations. Each of these locations will be sampled once per sampling event, for a total of 5 samples, plus additional QA/QC samples. The annual sampling locations are also presented on Figure 3.

The following section describes the methods that will be used for soil sample collection. All soil samples will be collected to a depth of about 6 inches. Prior to sample collection, the following field observations will be documented at each sampling location:

- Presence of surface water
- Visible staining
- Subsurface conditions

Soil samples will be collected in laboratory-supplied sample containers from the locations identified in Figure 3. Samples will be collected using either a stainless steel or disposable digging device (i.e. trowel, spoon) or an EnCore® sampler. Care will be taken to avoid cross-contaminating samples. Composite samples may be employed for submittal to the laboratory provided that no more than 10 grab samples are used to create a single composite sample. Composites should be made up of three to five grab samples. Composites may not be used for VOC analysis or EnCore® samples.

EnCore® samples will be collected with an EnCore® T-Handle sampler. Before taking the sample, prepare the sampler taking the following steps:

1. Hold coring body of EnCore® and push the plunger rod down until the small o-ring rests against the tabs.
2. Depress the locking lever on the EnCore® T-Handle.
3. Place coring body, plunger end first, into the open end of the T-Handle, aligning the slots on the coring body with the locking pins in the T-Handle.
4. Twist coring body clockwise to lock pins in the slots.
5. Ensure EnCore® sampler is locked in place.

Next collect a sample:

6. Turn the T-Handle to position the T up and the coring body down. The plunger bottom should now be flush with the bottom of the coring body.
7. Using the T-Handle, push the EnCore® sampler into the soil until the coring body is completely full. When it is full, the small o-ring will be centered in the T-Handle viewing hole.
8. Remove the EnCore® sampler from the soil.
9. Wipe any excess soil from the coring body exterior.
10. Cap the coring body while it is still in the T-Handle, by pushing the cap over the flat area of the ridge and twist to lock cap in place. Note: The cap must be seated to seal the sampler.

Remove sample and prepare for shipment:

11. Remove the capped EnCore® sampler by depressing the locking level on the T-Handle while twisting and pulling the EnCore® sampler from the T-Handle.
12. Lock the plunger by rotating the extended plunger rod fully counter clockwise until the wings rest firmly against the tabs.
13. Fill out the sample information on the back of the EnCore® sampler zipper bag.
14. Place the EnCore® sampler into the zipper bag, seal the bag, and place the sample on ice.

To collect the EnCore® sample, field personnel will dig to a depth of 2 – 6 inches using a clean stainless steel or disposable digging device. Upon exposing soil in the subsurface, the EnCore® sample will be immediately collected to avoid loss of VOCs in the sample. Three EnCore sample volumes are required per sample for laboratory analysis.

Soil samples for all other analyses will be collected in open mouthed glass jars, or other containers if directed by the contracted laboratory. Samples will be collected from 0 – 6 inches. Collection of surface soil can be accomplished with tools such as spades, shovels, and scoops. A stainless steel, plastic, or disposable scoop or spoon will suffice in most applications. Care should be exercised to avoid the use of devices plated with chrome or other materials. Plating is particularly common with garden trowels.

Follow these procedures to collect soil samples with a scoop or trowel:

1. Using a clean digging device remove vegetation and top layer of soil, then loosen the desired volume of soil from the sampling area.
2. Transfer the discrete grab sample into an appropriate laboratory sample container.
3. For a composite sample, homogenize grab samples in a stainless steel or glass-mixing container using a spoon or trowel.
4. Fill the sample container with the homogenized mixture.
5. Secure the cap tightly.

6. Label sample containers, and record appropriate data on sample data sheets (depth, location, color, and other observations).
7. Place glass sample containers into protective sleeves or bubble wrap, and place containers into an iced shipping container. Samples should be cooled to 4°C as soon as possible. It is suggested that the wrapped sample containers be placed in sealable plastic bags to reduce the chance of exposure to any ice melt, should it occur.
8. Complete chain of custody forms and ship as soon as possible to minimize sample-holding time. Scheduled arrival time at the analytical laboratory should give as much of a holding time as possible for scheduling of sample analysis.
9. Follow required decontamination procedures (see Section 6.7) if disposable equipment is not used.

6.6.2 Stormwater Runoff Samples – Water and Sediment

Stormwater runoff samples will be collected from two locations. These locations are labeled on Figure 3. The lines connected to points A and B on Figure 3 are storm sewer pipes. There is a small bed of rocks located at the outfall of each pipe. The two stormwater runoff locations will be sampled during both baseline and annual sampling events. Samples will include both stormwater runoff, if water is present, and runoff sediment during sampling events.

When sampling for both stormwater runoff and runoff sediment at the same location, the stormwater runoff sample should always be collected first. In collecting sediment samples, care must be taken to minimize disturbance and sample washing as it is retrieved through the liquid column above. Sediment fines may be carried out of the sample during collection if the liquid above is flowing or deep. This may result in collection of a non-representative sample due to the loss of contaminants associated with these fines. A sediment sample is usually expected to be a solid matrix; however sampling personnel should avoid placing the sample in the bottle and decanting off the excess liquid. Decantation promotes the loss of water-soluble compounds and volatile organics present in the sediment. If the sample is collected properly, any liquid that makes it into the bottle is representative of sediment conditions.

Stormwater Runoff Samples

Stormwater runoff samples from water flowing through a pipe into a ditch, receiving water, or basin should be collected directly from the pipe flow. For hard to reach pipes, it may be necessary to fasten a collection bottle to a pole. Alternately, for stormwater runoff in a ditch, swale, detention basin, or retention basin, stormwater runoff should be collected directly above sediments, near depositional areas where the water current is slower but consistently moving, and there is greater retention time for the runoff water to accumulate contaminants from sediment. A grab sample may be collected directly into the sample bottle from a consistently flowing part of the stormwater runoff stream.

For this project, it is anticipated that water may not be flowing from the storm drain pipe or even present during some sampling events. Water could be present seasonally or during rain events. If water is not present during a sampling event, stormwater runoff samples will not be collected. However, sediment samples will be collected. When water is present, samples will be collected as follows.

If water is flowing from the storm drain pipe, samples will be collected directly into sample containers from the pipe outflow. Fill the sample bottle directly from the discharge. If accessing the discharge point is problematic (i.e. access is too far or dangerous), use a pole or other appropriate sampling apparatus. The sample will be collected from a turbulent section in the central part of the flow. Fill the sample bottle nearly to the top by holding the opening into the flow of water. The bottles should not be rinsed or overfilled, this is especially true for bottles containing preservative.

For metals samples, filtering can be performed using a peristaltic pump and clean Tygon™ (or other non-reactive material) tubing equipped with a new in-line 0.45 µm filter. Place the tubing inlet in the water flow stream at the point of discharge. Let water flow through the tubing, and completely fill and exit the filter prior to beginning to fill the sample container. A new piece of clean tubing will be used for each sample location.

For VOC samples, once the sample vial has been filled to near the top, additional water maybe captured in the vial cap and added to the sample to form a convex meniscus at the top of the vial. This must be done immediately and quickly to avoid loss of organics. Alternately, if the flow is too great to capture the sample in the vial without overfilling or causing too much agitation, a peristaltic pump may be used to collect VOC samples, as described above for metals samples. However, VOC samples should not be collected through the 0.45 µm filter. Once completely filled, having formed a meniscus, carefully cap the vial, invert and tap it gently, and observe the vial for 10 seconds to ensure no air bubbles appear. If air bubbles are present, discard the sample and begin the VOC sample again using a new sample vial.

If water is not flowing from the pipe but is present in sufficient quantity, collect water samples from the ponded area. Field notes and records for these samples should clearly indicate the sample is not from flowing stormwater discharge and is from a ponded area. Taking care not to disturb any sediment in the ponded area, submerge the sample container and fill to nearly the top. For sample containers with preservative, and depending on the depth of the ponded area, options for filling preserved sample containers include: submerging only part of the container lip and avoiding over filling; filling a clean intermediate container such as a glass sample bottle and filling the sample container from it, or collecting samples using a peristaltic pump.

Alternately, it may be most appropriate to collect all samples from a ponded area at these sample locations using a peristaltic pump regardless of analysis or preservative. The peristaltic pump will be equipped with clean Tygon™ (or other non-reactive material) tubing. Place the inlet end into the ponded area taking care not to disturb the bottom or entrain sediment. Once the pump is on and water is flowing, samples can be collected into the appropriate sample container. For VOC samples, form a convex meniscus at the top of the vial, carefully cap the vial, invert and

tap it gently, and observe the vial for 10 seconds to ensure no air bubbles appear. If air bubbles are present, discard the sample and begin the VOC sample again using a new sample vial. For metals samples, attach a new 0.45 µm filter to the discharge line, and let water flow through the filter and exit prior to beginning sample collection. A new piece of clean Tygon™ tubing will be used for each sample location.

Runoff Sediment Samples

For this project, the stormwater runoff flow and depth, if present during the sampling events, are expected to be minimal and sediment should be easily reached. Under these conditions, a trowel or scoop may be used to collect the sediment. Generally, where the liquid above the sediment collection point is flowing or is greater than four inches deep, a corer or clamshell grab sampler should be used to collect the sample in an attempt to minimize washing the sediment as it is retrieved through the water column.

Sediment samples will be collected from the 0 – 6 inch interval near the pipe outfall area using a clean hand scoop. If deeper sediment samples are required, a core sampler or clamshell grab sampler should be used. Composite samples can be collected by mixing grab samples into a single sediment sample. Composites should consist of three to five grab samples, and never more than 10. Grab samples collected with a decontaminated or disposable sampling scoop will be placed in a decontaminated or disposable bowl or pan. The composite sample is then stirred until the texture and color appear uniform. Next scoop the composite sample into the sample containers and seal them. The sample containers are then labeled, wrapped in protective material, and placed in the shipping cooler and maintained at 4 °C.

6.6.3 Surface Water Samples

A total of four surface water samples, plus additional QA/QC samples, will be collected from four locations in proximity to the ARMD. Samples will be collected from the surface water locations shown in Figure 3. Samples will be collected during baseline and annual sampling events to provide PADEP with concentrations of the indicator contaminants described in Section 1.2.2.

The sampling locations were selected to represent the quality of surface water in the vicinity of the ARMD. Surface water sampling sites were selected by PADEP with concurrence from principal decision makers and are based on visits by PADEP to the ARMD site. However, these locations are subject to seasonal runoff. These surface water locations will be used to establish baseline concentrations of the potential contaminants before operation and are located in water channels that could be affected by the ARMD in the event of a release.

All surface water samples will be collected following the procedures outlined below. Before collecting samples, water body characteristics will be recorded in the field logbook and the following field parameters will be measured at each sampling location:

- Flow rate (if applicable)
- Dissolved oxygen
- Oxidation-reduction potential (ORP)
- Temperature
- Specific conductance
- pH
- Turbidity

Field parameters will be measured using an in-situ probe, such as a YSI or Horiba multiparameter water quality meter. At a minimum, the water quality meter will be calibrated at the start of each field day prior to use, and additionally as needed.

Surface water samples will be collected in laboratory-supplied sample containers. The surface water samples will be collected to avoid disturbing sediment from both the sides and bottom of the water body. The sampler will also avoid stagnant water. Stream samples will be collected as far from the bank and as close to the main current as possible.

Sampling personnel may encounter situations where rate of flow affects their ability to collect a sample. For fast flowing rivers and streams it may be nearly impossible to collect a mid-channel sample at a specific point. For low flowing shallow streams, the sampler should attempt to find a location where flow is naturally obstructed and a pool created which affords some depth from which to better submerge sample bottles. In no way should the environmental setting be altered with the intent to construct an artificial condition which aids in capturing a naturally occurring surface water sample.

It is anticipated that the surface water locations for this project will be shallow and narrow streams that do not require entry into the water channel. However, if entry is necessary, the field person will enter the channel downstream and walk upstream to minimize and avoid sediment disturbance. Once in the sampling location, the field person will stop and allow any disturbance to clear. Samples will be collected facing upstream and out in front of the field person.

Assuming the stream's water depth is shallow and the channel narrow, samples will be collected directly into sample containers by submerging the sample containers. It may be necessary to locate an area of the stream that has a naturally occurring pool of sufficient depth to submerge the container, or at a minimum enough of the container's opening to collect the sample. Preserved sample containers should be filled with care to avoid over sampling or loss of preservative. In the event a preserved container is over filled or preservative is lost, it should be discarded and sampled again with a new container. For VOC samples, once the sample vial has been filled to near the top, additional water maybe captured in the vial cap and added to the

sample to form a convex meniscus at the top of the vial. This must be done immediately and quickly to avoid loss of organics.

If the stream's water depth and flow is sufficient, the surface water samples will be collected from the centroid of the stream flow. Non-preserved samples will be collected directly into sample containers by submerging the sample containers and removing the lid. Collection of preserved samples in this manner is likely unfeasible without losing preservative or over sampling, so a clean intermediate container should be used. The intermediate container, such as a glass sample container, can be filled in the same manner as the non-preserved sample. The water is then immediately transferred to the preserved sample containers. For surface water VOC samples, form a convex meniscus at the top of the vial, carefully cap the vial, invert and tap it gently, and observe the vial for 10 seconds to ensure no air bubbles appear. If air bubbles are present, discard the sample and begin the VOC sample again using a new sample vial.

It should be noted that deep and wide streams require a sample representative of the cross-sectional area of the stream channel. This entails collecting grab samples across the channel and at various depths, combining them into a churn splitter, and filling sample containers from the spigot while churning. Because the streams for this project are assumed to be shallow and narrow this sample technique is not addressed in the ESP.

Surface water samples for metals will be field filtered with a 0.45 μm filter. Sample collection and filtering will be performed using a peristaltic pump and clean TygonTM (or other non-reactive material) tubing equipped with an in-line 0.45 μm filter. Submerge the tubing inlet into the stream flow, avoiding the stream bottom or any sediment. Turn the pump on. Let water flow through the tubing, and completely fill and exit the filter prior to beginning to fill the sample container. A new piece of clean tubing will be used for each sample location.

All surface water samples should be labeled, wrapped in protective material, and placed in the shipping cooler and maintained at 4 °C.

6.6.4 Groundwater Elevation

A least monthly, groundwater levels will be collected and recorded from each of the four monitoring wells depicted on Figure 3 for a period of 12 months. After the first year, groundwater level measurements will be collected during groundwater sampling events. Water levels from all four wells will be collected on the same day and as close together as possible. Measurements will be collected using a decontaminated water level indicator. Alternately, LEAD may install continuous water level dataloggers (such as a Solinst Levellogger) to record more frequent data. The download frequency will depend on the datalogger data capacity, but is expected to occur monthly.

Water levels will be collected using a specific reference point for each monitoring well, the established top of casing (TOC) measurement point. The TOC will have been surveyed for elevation to a resolution of 0.01 feet. Water levels will be converted as necessary to determine the elevation of groundwater, and taking the well stickup height into account. Groundwater

contours maps will be generated from the water level data and included in each sampling event report submitted to PADEP. The contour maps will be updated during each sampling event, or more frequently as needed from monthly water level data.

6.6.5 Groundwater Samples

Four monitoring wells will be sampled for baseline and annual sampling events at the locations depicted on Figure 3. One monitoring well will be located up gradient from the ARMD and three wells will be located down gradient of the ARMD. At the time of monitoring well installation, TOC measurement locations will be established and surveyed for elevation to a resolution of 0.01 feet. Additionally well construction and development logs will be created. This information will be used during sampling events. Specifics of the monitoring well installation can be found in the Weston well installation plan approved by PADEP in December 2015, see Attachment A.

Four groundwater samples and additional QA/QC samples, will be collected from the four monitoring wells depicted in Figure 3 during baseline sampling events. These sample results will be used to establish baseline conditions. The same four monitoring wells will be utilized for operational sampling events. Four groundwater samples, plus QA/QC samples, will be collected on a quarterly basis for the first year of operation of the ARMD, and then transition to an annual basis. Operational samples will provide PADEP with concentrations of the indicator contaminants described in Section 1.2.2, following the ARMD startup.

Groundwater sample collection will be conducted using low flow sampling techniques. Either a portable pump, such as a QED Sample Pro® MicroPurge® pump, or a dedicated pump such as a QED Well Wizard® will be used for low flow sampling. In either case, dedicated tubing will be employed. The pumps mentioned are non-contact bladder pumps, in which water enters a Teflon bladder, and then compressed gas is used to squeeze the bladder, forcing the water through the sample tubing line and out of the well. The compressed gas can be supplied in the form of a compressed gas cylinder or an air compressor. The gas pressure, and rate of discharge and refill are controlled by a control box, generally powered with batteries. The CONTRACTOR performing sampling will determine the actual pumps and accessories needed, and operate them following manufacturer instructions.

The depth to water (DTW) at all four groundwater monitoring wells will be measured at the start of the sampling event. DTW will be measured from the TOC at each monitoring well, in feet to a resolution of 0.01. The DTW measurements will be collected on the same day, as close together as possible, prior to any purging or sampling activities. The DTW measurements at all four wells will be performed together, regardless of when sampling actually occurs.

Upon arrival to the groundwater monitoring well at the time of sampling the following procedures will be performed.

1. Don appropriate personal protective equipment.
2. Confirm the monitoring well identification.

3. Calibrate field instruments in accordance with the manufacturer's directions. Record all calibration documentation in the field logbook or field form. Equipment calibration can be performed prior to arriving at the monitoring well, but must be performed once each day before sampling begins.
4. Measure the depth to static water level and the total well depth to the nearest 0.01 feet from the TOC measurement point using an electronic water level indicator. Record the water level measurement and total depth in the field logbook or on the field form.
5. Consult the well construction log to determine the well's screen interval. **Be sure to take into account the point of reference, below ground surface (bgs) or below TOC.**
6. If the screen is completely below the groundwater level, the pump inlet will be placed in the middle of the screen interval.
7. If the screen is not completely below the groundwater level, the pump inlet will be placed in the middle of the "wetted screen" interval.
8. To determine the wetted screen interval and the pump inlet depth, subtract the DTW from the depth to the bottom of the screen, then divide by 2, and add the result to the DTW.

For example, a well screened from 10 to 35 feet bgs with a water level of 17 feet bgs will have a wetted screen interval between 17 and 35 feet bgs and the pump inlet should be placed at 26 feet bgs.

Solve for pump inlet depth (C) using the following equation:

$$[(A - B) \div 2] + B = C$$

Where:

A = Bottom screen depth (35 feet bgs)

B = Depth to water (17 feet bgs)

C = Pump inlet depth

9. Carefully lower the pump into the well. Each well will have dedicated tubing for sampling. Alternately, a dedicated pump and tubing may be used and is recommended.
10. Place the inlet to the pump at the middle of the wetted screen.
11. The pump speed should be set so that the water column in the well does not continually drop during purging. The pump rate should be set so that a constant water level is maintained with minimal drawdown.
12. Monitor water quality parameters every 5 minutes during purging using an in-line meter and flow through cell (such as a Horiba U-series). Record the water quality parameters on the groundwater sampling log form. If the water quality parameters are stable for three consecutive readings, collect samples for chemical analysis.

The following field parameters, measured in a flow through cell (with the exception of turbidity), will be used to indicate stability has been achieved. Depth to water will also be measured, recorded, and used to indicate recharge water is being collected.

- Dissolved oxygen ($\pm 10\%$ for values greater than 0.5 mg/L, if three DO values are less than 0.5 mg/L the parameter is considered stable)
- ORP (± 10 mV)
- Temperature ($\pm 1^\circ\text{C}$)

- Specific conductance ($\pm 3\%$)
- pH (± 0.1)
- Turbidity ($\pm 10\%$ for values greater than 5 NTU, if three readings are less than 5 NTU the parameter is considered stable)
- Water drawdown (≤ 0.3 feet)

If the water quality parameters have not stabilized, continue purging until stabilization occurs. If stable parameters are not obtained within three successive readings at 5-minute intervals, the well will be purged for a maximum of 8 readings or 40 minutes prior to collecting the groundwater sample. Achievement of turbidity levels of less than 5 NTU, and stable drawdowns of less than 0.3 feet, while desirable, are not mandatory.

Turbidity samples are collected before the flow-through-cell. A "T" connector coupled with a valve is connected between the pump's tubing and the flow through cell. When a turbidity measurement is required, the valve is opened to allow the groundwater to flow into a container. The valve is closed and the container sample is then placed in the turbidity meter. NTU calibration standards for the turbidity meter are required to obtain accurate and valid turbidity readings.

13. Reduce the pump flow to a rate of less than 300 mL per minute, and collect samples, the rate should be reduced to 100 mL per minute for the collection of VOC samples.

Note: Do not stop the pump after stabilization and prior to sample collection.

14. Collect samples from the discharge of the pump – DO NOT SAMPLE THROUGH THE FLOW THROUGH CELL OR "T" CONNECTOR. Fill the laboratory supplied sample containers, collecting any necessary field QC samples required, in the same manner in which the samples were collected. Duplicates should be collected together, filling each container type consecutively, before filling the next container type. Special handling for some analyses is discussed below.
15. Label, package, and prepare the samples for shipment to the laboratory.
16. Transfer the samples to cold storage (i.e. a cooler) after collection and maintain them at 4 °C.

Special handling for collection of dissolved metals and VOCs is required. Dissolved metals will be field filtered with a disposable, inline, 0.45µm filter connected to the sample collection tubing. VOC samples will be collected in pre-preserved, amber, 40 mL VOA glass vials following these instructions:

1. Open the vial, set cap in a clean place, and collect the sample during the middle of the cycle. When collecting duplicates, collect both samples at the same time.
2. Fill the vial to just overflowing. Do not rinse the vial, nor excessively overfill it. There should be a convex meniscus on the top of the vial.
3. Check that the cap has not been contaminated from splashing and carefully cap the vial. Place the cap directly over the top and screw down firmly. Do not over tighten and break the cap.

4. Invert the vial and tap gently. Observe vial for at least 10 seconds. If an air bubble appears, discard the sample and begin again. It is imperative that no entrapped air is in the sample vial.
5. Place the vial in protective foam or bubble wrap material and place into the cooler, oriented so that it is lying on its side, not straight up.
6. Ensure that the samples remain at 4 °C, but do not allow them to freeze.
7. The holding time for VOCs is 7 days. If preserved with HCl to a pH < 2, as planned for this project, the holding time is 14 days. Three 40 mL VOA containers are required for each sample.

Groundwater samples will be collected in laboratory-supplied sample containers from each of the four wells identified in Figure 3. Samples will not be collected through the flow through cell. Care will be taken to avoid cross-contaminating samples. Dedicated tubing will be used for each well, and preferably a dedicated pump. Decontamination of sampling equipment is therefore not required and associated rinsate samples will not be collected. However, if dedicated pumps are not utilized, decontamination of the pump will be required and rinsate samples collected. Decontamination procedures and rinsate samples are discussed in Section 6.7 and Section 6.6.6 of this document, respectively.

6.6.6 Collection of Quality Assurance and Quality Control Samples

Quality assurance and quality control samples are discussed in Section 6. This section addresses how QA/QC samples are collected in the field. Expected and unanticipated QA/QC samples include duplicates, trip blanks, temperature blanks, equipment rinsate blanks, source water blanks, and matrix spike/matrix spike duplicates (MS/MSD).

Collection of duplicate samples provides for the evaluation of the laboratory's and field sampling team's performance by comparing analytical results of two samples from the same location. Duplicate samples will be collected from each matrix at a rate of 10 percent and submitted to the lab as "blind" samples.

Duplicates of water samples (groundwater, surface water, and runoff water) will be obtained at the same time the environmental sample is collected. The duplicate sample will be obtained by alternately filling sample containers from the same sampling device for each parameter. For example, groundwater samples and duplicate samples will be collected from the pump discharge line, filling first the sample and then the duplicate container for inorganics, next the sample and then the duplicate container for perchlorate, etc. The vials for VOCs will be alternately filled for sample then duplicate until all 6 vials have been filled. The duplicate will be handled in an identical manner to the sample, including the volume collected, containers filled, preservatives used and holding times. Samples will be placed in protective sleeves or bubble wrap as needed (i.e. glass), placed in the shipping cooler and cooled to 4°C as soon as possible. Duplicate sample information will be entered onto the COC and duplicate samples shipped to the laboratory with the other environmental samples.

Obtaining duplicate samples in a soil or sediment matrix requires homogenization of the sample prior to filling sample containers. Regardless, volatile organic samples must always be taken from discrete locations or intervals without compositing or mixing. This practice is necessary to prevent loss of volatile constituents and to preserve, to the extent practicable, the physical integrity of the volatile fraction. Homogenization of the sample for remaining parameters is necessary to generate two equally representative samples. Note that enough sample must be collected at one time in order to fill all the necessary sample containers. It may be necessary to co-locate or depth-integrate collection so enough sample volume is available.

Homogenization and collection of the soil and sediment samples will be accomplished by filling a properly decontaminated stainless steel tray or bowl with the sample and mixing it with a decontaminated stainless steel or Teflon® spoon. The extent of mixing required will depend on the nature of the sample and should be done to achieve a consistent physical appearance prior to filling sample containers. Once mixing is completed the sample should be divided in half and sample containers should be filled by alternately scooping material from each half into the sample container. The duplicate will be handled in an identical manner to the environmental soil and sediment samples, including the volume collected, containers filled, and holding times. Samples will be placed in protective sleeves or bubble wrap, placed in the shipping cooler and cooled to 4°C as soon as possible. Duplicate sample information will be entered onto the COC and duplicate samples shipped to the laboratory with the other environmental samples.

A trip blank is a container of laboratory reagent water that is shipped, unopened, to and from the field, with empty and full sample containers. The laboratory prepares trip blanks. There are no collection requirements for trip blanks. Instead field personnel must ensure trip blanks accompany each shipping cooler that contains aqueous VOC samples. Additionally, trip blanks must not be opened at any time once prepared by the lab until the trip blank is returned to the lab for analysis. It is imperative that field personnel record each trip blank with its associated samples. Like the trip blank, the temperature blank is provided by the laboratory and does not require collection in the field. Field personnel ensure a temperature blank is included in each cooler shipment of environmental samples sent to the laboratory.

A rinsate blank is a sample collected after a sampling device is subjected to standard decontamination procedures. Reusable equipment that has been decontaminated is assumed to be clean and not suspected of contaminating samples. Rinsate samples are collected to confirm this assumption. To collect a rinsate sample, distilled/deionized water will be poured over or through the sampling equipment into sample containers. The rinsate blanks will be handled in an identical manner to the environmental aqueous samples, including the volume collected, containers filled, preservatives used and holding times. Samples will be placed in protective sleeves or bubble wrap as needed (i.e. glass), placed in the shipping cooler and cooled to 4°C as soon as possible. Rinsate blank sample information will be entered onto the COC and samples shipped to the laboratory with the other environmental samples. It is imperative that field personnel associate and record each sample collected using the decontaminated equipment and the associated rinsate blank for that equipment.

Source-water blanks verify that the water used to acquire the equipment rinsate blank was analyte-free prior to collecting the rinsate blank. The source-water blank is collected by pouring the source-water directly into the sample containers for aqueous sample analysis. The source-water blanks will be handled in an identical manner to the samples, including the volume collected, containers filled, preservatives used and holding times. Samples will be placed in protective sleeves or bubble wrap as needed (i.e. glass), placed in the shipping cooler and cooled to 4°C as soon as possible. Source-water blank sample information will be entered onto the COC and source-water blank samples shipped to the laboratory with the other environmental samples.

MS/MSD samples are environmental samples that are spiked in the laboratory to verify percent recoveries. MS/MSD samples do not require collection methods different than those of the environmental samples, and do not even require a different sample ID. However, laboratories generally require a greater sample volume to perform the laboratory QA/QC analyses. The contracted laboratory for this project will provide the CONTRACTOR with the required sample volume needed for MS/MSD samples. The COC will clearly mark an environmental sample as a MS/MSD sample and indicate the additional volume.

6.7 SAMPLING EQUIPMENT DECONTAMINATION PROCEDURES

An important aspect of quality control is the decontamination of field sampling equipment. Improperly cleaned and prepared sampling equipment can lead to misinterpretation of environmental data due to interference caused by cross-contamination.

Decontamination of non-disposable sampling equipment will be performed to prevent the introduction of extraneous material into samples and to prevent cross-contamination between samples. All sampling equipment will be decontaminated by washing with a non-phosphate detergent such as Liquinox™ or equivalent.

It is anticipated that soil and sediment sampling equipment (i.e. trowels and bowls) may not always be disposable, and that decontamination will be required. However, though new bladders and dedicated tubing will be used in conjunction with pumps for groundwater sampling, if well-specific dedicated pumps are not utilized the other pump components must be decontaminated. This will require disassembly of the pump for proper cleaning. All interior and exterior surfaces must be scrubbed and rinsed as directed below, including water inlet orifices.

The following steps will be followed for decontamination of non-disposable sample equipment.

1. Rinse with potable water — This step will decrease the gross contamination and reduce the frequency at which the non-phosphate detergent and water solution needs to be changed. A 5-gallon bucket about 75 percent full of water and a long-handled brush is suggested for this step. Frequent changing of this water will increase its effectiveness.
2. Wash with non-phosphate detergent and water solution — This step will remove all visible contamination from the equipment. A 5-gallon bucket approximately 75 percent

full of solution and a long-handled brush is suggested for this step. Dilute the non-phosphate detergent as directed by the manufacturer.

3. Rinse with potable water — This step will rinse all the detergent solution away from the equipment. A 5-gallon bucket approximately 75 percent full of water and a long-handled brush is suggested for this step. Periodic changing of this water is required.
4. Rinse with distilled or deionized water — This step will rinse any detergent solution and potable water residues. Rinsing is most effective by pouring the distilled or deionized water from the water container over the equipment over a 5-gallon bucket. Applying the distilled or deionized water using a Nalgene™ squeeze bottle or sprayer while holding equipment over a 5-gallon bucket is another option.

6.8 SAMPLE HANDLING AND CUSTODY

The following sections present the requirements for sample handling and custody. Sample container type, required sample volume, preservatives, and holding times for samples prior to extraction and analysis were previously summarized in Section 5.4, Table 5.

6.8.1 Sample Identification

A unique sample ID number will be assigned to each sample collected during this project. The sample ID numbering system is designed to be compatible with a computerized data management system that may include previous results for samples collected at the ARMD. The sample numbering system allows each sample to be uniquely identified and provides a means for tracking the sample from collection through analysis. The numbering system indicates sample location, date of sample, and QA/QC sample type if applicable. The suggested sample identification nomenclature for ARMD, and shown in Table 7, includes the following. The CONTRACTOR may choose a different sample ID method, however the method will be consistent for all sampling events at the ARMD.

Facility Identification	ARMD
Location Identifier	Monitoring well 1 = MW1, etc. Surface water location 1 = SW1, etc. Soil location 1 = SL01, etc Surface runoff location A = A
For Surface Runoff Type	RW = runoff water RS = runoff sediment
Date	Day, month, year (8 digits) = 05142017
If applicable, QA/QC Sample Type	D = duplicate

Field QC samples for this investigation may include field duplicates, equipment rinsates, source water blanks, trip blanks, and temperature blanks. Field duplicates will be identified by adding a “D” to the end of the sample identification for the primary sample collected from a specific sample location, these have been defined in Table 7.

Equipment rinsates, source water blanks, and trip blanks will have sample IDs beginning with ARMD, the corresponding date, followed by “ER,” “SB,” or “TB,” respectively, and then a sequential number. An example ID for the first equipment rinsate collected on June 14, 2017 during a sampling event would be ARMD06142017ER01. If an additional equipment rinsate sample was collected that day the ID would be ARMD06142017ER02.

6.8.2 Sample Labels

A sample label will be affixed to each sample container. The label will be completed with the following information, written in indelible ink:

- Project name and location
- Sample ID number
- Date and time of sample collection
- Preservative used
- Sample collector’s initials
- Analysis required

After labeling, each sample will be refrigerated or placed in a cooler that contains ice to maintain the sample temperature at 4 ± 2 °C.

6.8.3 Sample Documentation

Documentation during sampling is essential to ensure proper sample identification. CONTRACTOR personnel will adhere to the following general guidelines for maintaining field documentation:

- Documentation will be completed in permanent black ink.
- All entries will be legible.
- Errors will be corrected by crossing out with a single line and then dating and initialing the lineout.
- Any serialized documents will be maintained at a CONTRACTOR office and referenced in the site logbook.
- Unused portions of pages will be crossed out, and each page will be signed and dated.

Section 4.5 addresses how CONTRACTOR will use logbooks to document field activities. The CONTRACTOR field team leader is responsible for ensuring that sampling activities are properly documented.

6.8.4 Chain-of-Custody

CONTRACTOR will use standard sample custody procedures to maintain and document sample integrity during collection, transportation, storage, and analysis. A sample will be considered to be in custody if one of the following statements applies:

- It is in a person's physical possession or view
- It is in a secure area with restricted access
- It is placed in a container and secured with an official seal such that the sample cannot be reached without breaking the seal

Chain-of-custody procedures provide an accurate written record that traces the possession of individual samples from the time of collection in the field to the time of acceptance at the laboratory. The chain-of-custody record will also be used to document all samples collected and the analysis requested. Information that the field personnel will record on the chain-of-custody record includes:

- Project name and number
- Sampling location
- Name and signature of sampler
- Destination of samples (laboratory name)
- Sample identification number
- Date and time of collection
- Number and type of containers filled
- Analysis requested
- Preservatives used (if applicable)
- Filtering (if applicable)
- Sample designation (grab or composite)
- Signatures of individuals involved in custody transfer, including the date and time of transfer
- Air bill number (if applicable)
- Project contact and phone number

Unused lines on the chain-of-custody form will be crossed out. Field personnel will sign chain-of-custody records that are initiated in the field, and the air bill number will be recorded on the form. The record will be placed in a waterproof plastic bag and taped to the inside of the shipping container used to transport the samples. Signed air bills will serve as evidence of custody transfer between field personnel and the courier, and between the courier and the laboratory. Copies of the chain-of-custody record and the air bill will be retained and filed by field personnel before the containers are shipped.

Laboratory chain-of-custody begins when samples are received and continues until samples are discarded. The laboratory should designate a specific individual as the sample custodian. The custodian will receive all incoming samples, sign the accompanying custody forms, and retain copies of the forms as permanent records. The laboratory sample custodian will record all pertinent information concerning the samples, including the persons delivering the samples, the date and time received, sample condition at the time of receipt (sealed, unsealed, or broken container; temperature; or other relevant remarks), the sample ID numbers, and any unique laboratory identification numbers for the samples. This information should be entered into a computerized laboratory information management system. When the sample transfer process is

complete, the custodian is responsible for maintaining internal logbooks, tracking reports, and other records necessary to maintain custody throughout sample preparation and analysis.

The laboratory will provide a secure storage area for all samples. Access to this area will be restricted to authorized personnel. The custodian will ensure that samples requiring special handling, including samples that are heat- or light-sensitive, or have other unusual physical characteristics, will be properly stored and maintained prior to analysis.

6.8.5 Sample Shipment

The following procedures will be implemented when samples collected during this project are shipped:

- The cooler will be filled with bubble wrap, sample bottles, and packing material. Sufficient packing material will be used to prevent sample containers from breaking during shipment. Enough ice will be added to maintain the sample temperature below 4 ± 2 °C.
- The chain-of-custody records will be placed inside a plastic bag. The bag will be sealed and taped to the inside of the cooler lid. The air bill, if required, will be filled out before the samples are handed over to the carrier. The laboratory will be notified if the sampler suspects that the sample contains any substance that would require laboratory personnel to take safety precautions.
- The cooler will be closed and taped shut with strapping tape around both ends. If the cooler has a drain, it will be taped shut both inside and outside of the cooler.
- Signed and dated custody seals will be placed on the front and side of each cooler. Wide clear tape will be placed over the seals to prevent accidental breakage.
- The chain-of-custody record will be transported within the taped, sealed cooler. When the cooler is received at the analytical laboratory, laboratory personnel will open the cooler and sign the chain-of-custody record to document transfer of samples.

Multiple coolers may be sent in one shipment to the laboratory. The outside of the coolers will be marked to indicate the number of coolers in the shipment.

6.9 EQUIPMENT TESTING, INSPECTION, AND MAINTENANCE

This section outlines the testing, inspection, and maintenance procedures that will be used to keep field equipment in good working condition.

6.9.1 Calibrating Field Equipment

Field equipment will be calibrated at the beginning of the field effort and at prescribed intervals. The calibration frequency depends on the type and stability of equipment, the intended use of the equipment, and the recommendations by the manufacturer. At a minimum, field equipment will be calibrated at the beginning of each field day prior to starting work. Detailed calibration procedures for field equipment are available from the specific manufacturers' instruction manuals, and general guidelines should be included in CONTRACTOR standard operating procedures (SOP).

All calibration information will be recorded in a field logbook or on field forms. A label that specifies the scheduled date of the next calibration will be attached to the field equipment. If this type of identification is not feasible, equipment calibration records will be readily available for reference.

6.9.2 Maintaining Field Equipment

Preventive maintenance for most field equipment is carried out in accordance with procedures and schedules recommended in (1) the equipment manufacturer's literature or operating manual or (2) SOPs that describe equipment operation associated with particular applications of the instrument. However, more stringent testing, inspection, and maintenance procedures and schedules may be required when field equipment is used to make critical measurements.

A field instrument that is out of order will be segregated, clearly marked, and not used until it is repaired. The field team leader (FTL) will be notified of equipment malfunctions so that service can be completed quickly or substitute equipment can be obtained. When the condition of equipment is suspect, unscheduled testing, inspection, and maintenance should be conducted. Any significant problems with field equipment will be reported in a daily field QC report.

6.9.3 Inspection and Acceptance of Supplies and Consumables

CONTRACTOR project managers have primary responsibility for identifying the types and quantities of supplies and consumables needed to complete the ARMD sampling and are responsible for determining acceptance criteria for these items.

Supplies and consumables can be received either at the CONTRACTOR office or at a work site. When supplies are received at an office, the project manager will sort them according to vendor, check packing slips against purchase orders, and inspect the condition of all supplies before they are accepted for use on a project. If an item does not meet the acceptance criteria, deficiencies will be noted on the packing slip and purchase order and the item will then be returned to the vendor for replacement or repair.

Procedures for receiving supplies and consumables in the field are similar. When supplies are received, the CONTRACTOR project manager will inspect the items against the acceptance

criteria. Any deficiencies or problems will be noted in the field logbook and deficient items will be returned for immediate replacement.

Analytical laboratories are required to provide containers certified as clean for all analyses. These containers must meet EPA standards described in “Specifications and Guidance for Obtaining Contaminant-Free Sampling Containers” (EPA 1992).

7.0 QUALITY ASSURANCE AND QUALITY CONTROL

CONTRACTOR will assess the quality of field data through regular collection and analysis of field QC samples. Field QC samples are collected and analyzed to check sampling and analytical precision, accuracy, and representativeness. Laboratory QC samples will also be analyzed in accordance with referenced analytical method protocols to ensure that laboratory procedures are conducted properly and that the quality of the data is known. Table 8 presents the minimum frequency of collection for precision and accuracy QC samples.

TABLE 8
QUALITY CONTROL SAMPLES FOR PRECISION AND ACCURACY

QC Type	Precision	Accuracy	Minimum Frequency
Field QC	Field Duplicates		10% - 1 per 10 samples for each sample media
		Trip blank	1 per ice chest used for VOC sample shipment or as required by laboratory
		Rinsate blank	10% per day per non-disposable sampling equipment
		Source-water blank	1 per source of water used for final equipment rinse during decontamination
Laboratory QC	MS/MSD RPD	MS/MSD Percent Recovery	5% - 1 per 20 samples for each sample media
		Method Blanks	5% - 1 per 20 samples for each sample media
		LCS or Blank Spikes	5% - 1 per 20 samples for each sample media

Notes:

GC Gas chromatography

GC/MS Gas chromatography/mass spectrometry

LCS Laboratory control sample

MS/MSD Matrix spike/matrix spike duplicate

QC Quality control

RPD Relative percent difference

Equipment rinsate samples and source-water blank samples are not anticipated because dedicated and disposable sampling equipment is intended for use during sampling events.

7.1 QUALITY ASSURANCE AND QUALITY CONTROL DEFINITIONS

7.1.1 Precision

Precision is the degree of mutual agreement between individual measurements of the same property under similar conditions. Usually, combined field and laboratory precision is evaluated by collecting and analyzing field duplicates and then calculating the variance between the

$$RPD = \frac{|A - B|}{(A + B)/2} \times 100\%$$

samples, typically as a relative percent difference (RPD).

where:

A = First duplicate concentration
B = Second duplicate concentration

Field sampling precision is evaluated by analyzing field duplicate samples. For this project, field duplicates will be collected at a minimum ratio of 10 percent, one duplicate for every 10 field samples.

Laboratory analytical precision is evaluated by analyzing laboratory duplicates or matrix spikes (MS) and matrix spike duplicates (MSD). For this project, MS/MSD samples will be generated for each analyte. The analytical results for each MS/MSD pair will be used to calculate an RPD for evaluating precision.

7.1.2 Accuracy

A program of sample spiking will be conducted by the laboratory to evaluate laboratory accuracy. This program includes analyzing the MS and MSD samples, laboratory control samples (LCS) or blank spikes, surrogate standards, and method blanks. The MS and MSD samples will be prepared and analyzed at a minimum frequency of 5 percent, each, for sample type (surface water, groundwater, and soil samples) collected in the field. The LCS or blank spikes are also analyzed at a minimum frequency of 5 percent. The results of the spiked samples are used to calculate the percent recovery for evaluating accuracy.

$$\text{Percent Recovery} = \frac{S - C}{T} \times 100$$

where:

S = Measured spike sample concentration

C	=	Sample concentration
T	=	True or actual concentration of the spike

7.1.3 Representativeness

Representativeness expresses the degree to which sample data accurately and precisely represent the characteristics of a population, variations in a parameter at a sampling point, or an environmental condition that they are intended to represent. For this project, representative data will be obtained through careful selection of sampling locations and analytical parameters. Representative data will also be obtained through properly collecting and handling samples to avoid interference and minimize contamination.

Representativeness of data will also be ensured through the consistent application of established field and laboratory procedures. Laboratory blank samples will be evaluated for the presence of contaminants to aid in evaluating the representativeness of sample results. Data determined to be non-representative, by comparison with existing data, will be used only if accompanied by appropriate qualifiers and limits of uncertainty.

7.1.4 Completeness

Completeness is a measure of the percentage of project-specific data that are valid. Valid data are obtained when samples are collected and analyzed in accordance with procedures outlined in this ESP, and when none of the criteria that affect data usability are exceeded. When the data validation is completed, the percent completeness value will be calculated by dividing the number of usable sample results by the total number of sample results planned for this investigation.

7.1.5 Comparability

Comparability expresses the confidence with which one data set can be compared with another. Comparability of data will be achieved by consistently following standard field and laboratory procedures and by using standard measurement units in reporting analytical data for both baseline and annual sampling.

7.1.6 Method Detection and Quantitation Limits

The method detection limit (MDL) is the minimum concentration of a compound that can be measured and reported. The MDL is a specified limit at which there is 99-percent confidence that the concentration of the analyte is greater than zero. The MDL takes into account sample matrix and preparation. The subcontractor laboratory will demonstrate the MDLs for all analyses.

The Practical Quantitation Limit (PQL) represents the lowest concentration of an analyte that a laboratory can accurately and reproducibly quantify with confidence in a sample matrix. The PQL takes into account changes in the preparation and analytical methodology that may alter the ability to detect an analyte, including changes such as use of a smaller sample aliquot or dilution of the sample extract. Physical characteristics such as sample matrix that may alter the ability to detect the analyte are also considered. PQLs are normally defined as 5 or 10 times the corresponding MDL. A laboratory reporting limit (RL) is the concentration in the sample that is equivalent to the lowest standard routinely run for that analyte by a given method and instrument combination. The laboratory sets the RL high enough so that laboratory instrument sensitivity does not exceed it.

7.2 QUALITY ASSURANCE AND QUALITY CONTROL FIELD SAMPLES

Field QA/QC samples are collected and analyzed to assess the quality of data. The baseline and annual sampling events will include collection and analysis of the QA/QC samples presented in this section. At the discretion of the principal decision makers or at the suggestion of the analytical laboratory, additional QA/QC samples may be collected.

Temperature blanks will be provided in each shipment container. Trip blanks will accompany each shipping cooler that contains aqueous VOC samples. Field duplicate samples will be submitted during the sampling event by collecting additional sample volumes and designating the sample containers for the additional volume with a separate sample ID number for analysis. An equipment rinsate blank will be collected daily from each type of non-disposable sampling equipment used during the sampling event. One source-water blank will be necessary for each source of water used for decontamination.

7.2.1 Temperature Blanks

A temperature blank is used to verify the temperature of the sample cooler upon receipt at the analytical laboratory. A temperature blank will be placed in each shipping cooler containing environmental or QC samples. The temperature blank will be opened upon receipt at the laboratory and the recorded temperature will document the temperature of the sample cooler.

7.2.2 Trip blanks

A trip blank is a container of laboratory reagent water that is shipped, unopened, to and from the field, with empty and full sample containers. Trip blanks are shipped with aqueous VOC samples. Their purpose is to identify contaminants introduced into samples during transit to and from the laboratory. Trip blanks will accompany each shipping cooler that contains aqueous VOC samples.

7.2.3 Duplicates

Duplicates are field samples obtained from one location. They are divided into separate sample containers. For soil samples they are homogenized with thorough hand mixing prior to being

divided into separate sample containers. However, VOC soil samples are not homogenized, but rather collected from a collocated discrete sample. Duplicates are treated as separate samples throughout the remaining sample handling and analytical processes.

Duplicate samples are used to assess total error (precision) associated with sample heterogeneity, sample methodology, and analytical procedures. This procedure is useful in determining total (sampling and analytical) error because it evaluates sample collection, sample preparation, and analytical procedures. Duplicates may be especially important when determining precision for critical samples with contamination concentrations near or above an action level.

7.2.4 Rinsate Blanks

Equipment rinsate blanks are generally collected during sampling at a frequency of once per day of sampling per team per type of tool used. For this project, when dedicated sampling equipment is not utilized, rinsate blanks per type of sampling device used will be collected at a rate of 10% per day. Rinsate blanks are samples obtained by running distilled/deionized water over decontaminated sampling equipment to test for residual contamination. The rinsate blank water is collected in sample containers for handling, shipment, and analysis. These samples are treated identical to the samples collected that day. A rinsate blank is used to assess cross contamination brought about by improper decontamination procedures. During data validation, the results of the rinsate blanks will be used to qualify data or to evaluate the levels of analytes in the field samples collected during the sampling event.

7.2.5 Source-Water blanks

One source-water blank will be collected from each source of water (distilled or deionized) used to collect the equipment rinsates. The source-water blank samples will verify that the water used to acquire the equipment rinsate was analyte-free prior to collecting the equipment rinsate.

7.3 LABORATORY QUALITY ASSURANCE AND QUALITY CONTROL

The subcontracted laboratory analyzing the samples collected during baseline and annual sampling events will implement and maintain its own QA/QC procedures. Subcontractor laboratories will prepare and follow a maintenance and calibration schedule for each instrument used to analyze samples collected for this project. All instruments will be serviced and calibrated at scheduled intervals necessary to optimize factory specifications and QA/QC requirements. Maintenance and calibrations will be documented according to the laboratory's QA/QC program. Only PADEP accredited laboratories maybe contracted and the laboratory's QA/QC program documentation will be supplied to LEAD upon request. Laboratory QA/QC procedures generally include:

- Matrix spikes
- Matrix spike duplicates
- Duplicates

- Blanks (field, trip, method, reagent instrument, decontamination, and source water)
- Internal standards and surrogates
- Calibration and calibration verification
- Control charts
- Standards and standard sources
- Reagents and gases

Matrix spike and matrix spike duplicate samples (MS/MSD) are environmental samples that are spiked in the laboratory or in the field with a known concentration of a target analyte(s) to verify percent recoveries. MS/MSD samples are primarily used to check sample matrix interferences. They can also be used to monitor laboratory performance; MS/MSD samples are used to evaluate error due to laboratory bias and precision.

8.0 SPECIAL TRAINING AND CERTIFICATION

This section outlines the anticipated training and certification required to complete the activities described in this ESP. The following sections describe the anticipated requirements for CONTRACTOR and subcontractor personnel, as applicable, that are working on site. A project-specific health and safety plan should be prepared prior to work.

8.1 HEALTH AND SAFETY TRAINING

CONTRACTOR personnel who work at hazardous waste project sites are required to meet the Occupational Safety and Health Administration (OSHA) training requirements defined in Title 29 *Code of Federal Regulations* (CFR) Part 1910.120(e). These requirements include: (1) 40 hours of formal off-site instruction; (2) a minimum of 3 days of actual on-site field experience under the supervision of a trained and experienced field supervisor; and (3) 8 hours of annual refresher training. Field personnel who directly supervise employees engaged in hazardous waste operations also receive at least 8 additional hours of specialized supervisor training. The supervisor training covers health and safety program requirements as well as training requirements, personal protective equipment (PPE) requirements, spill containment program, and health-hazard monitoring procedures and techniques. At least one member of every CONTRACTOR field team will maintain current certification in the American Red Cross “Multimedia First Aid” and “Cardiopulmonary Resuscitation (CPR) Modular,” or equivalent.

Copies of CONTRACTOR’s health and safety training records, including course completion certifications for the initial and refresher health and safety training, specialized supervisor training, and first aid and CPR training, should be maintained in project files.

Before work begins at a specific hazardous waste project site, CONTRACTOR personnel are required to undergo site-specific training that thoroughly covers and identifies the following:

- Names of personnel and alternates responsible for health and safety at a hazardous waste project site

- Health and safety hazards present on site
- Selection of the appropriate personal protection levels
- Correct use of PPE
- Work practices to minimize risks from hazards
- Safe use of engineering controls and equipment on site
- Medical surveillance requirements, including recognition of symptoms and signs that might indicate overexposure to hazardous substances
- Contents of the ARMD Health and Safety Plan (HASP) and the CONTRACTOR project-specific HASP.

8.2 SUBCONTRACTOR TRAINING

Subcontractors who work on site will certify that their employees have been trained for work on hazardous waste project sites. Training will meet OSHA requirements defined in 29 CFR 1910.120(e). Before work begins at the ARMD site, subcontractors will submit copies of the training certification for each employee to CONTRACTOR.

Employees of associate and professional services firms and technical services subcontractors will attend a safety briefing and complete a tailgate safety meeting form before they conduct on-site work. This briefing covers the topics described in Section 7.1 and is conducted by the CONTRACTOR on-site health and safety officer (OHSO) or other qualified person.

Subcontractors are responsible for conducting their own safety briefings. CONTRACTOR personnel may audit these briefings.

8.3 SPECIALIZED TRAINING AND CERTIFICATION REQUIREMENTS

Other than the health and safety training discussed above, no specialized training is anticipated for the soil, surface water, groundwater, stormwater runoff, and sediment runoff monitoring investigation. However, it is assumed that the project team will have existing field sampling experience.

9.0 DATA MANAGEMENT

Field and analytical data collected during this project, and other environmental investigations at the ARMD characterize site conditions and provide information which can support actions to protect human health and the environment. Data management will be handled to ensure efficient access to data so that decisions based on the data may be made in a timely manner.

This section describes the procedures that are planned to handle, review, verify, and validate field and laboratory data. These procedures ensure that the data are sufficient to meet PADEP requirements for the project.

The analytical results of samples collected for this project will be reported as an electronic data deliverable (EDD). Hardcopies may also be issued. Field, laboratory, and cartographic data collected from or generated during sampling events, as well as databases and supporting software (if applicable), will be archived and maintained in a secure location in the LEAD Environmental Office.

Project team personnel will verify field data by reviewing data sets to identify inconsistencies or anomalous values. Inconsistencies discovered will be resolved as soon as possible by seeking clarification from field personnel responsible for data collection. Field personnel will be responsible for following the sampling and documentation procedures described in this ESP so that defensible and justifiable data are obtained.

Data values that are significantly different from the population are called “outliers.” A systematic effort will be made to identify outliers or errors before field personnel report the data. Outliers can result from improper sampling or measurement methodology, data transcription errors, calculation errors, or natural causes. Outliers that result from errors found during data verification will be identified and corrected. Outliers that cannot be attributed to errors in sampling, measurement, transcription, or calculation will be clearly identified in project reports. Outlier samples will be resampled within 10 days following validation of the original sample result.

Laboratory personnel will verify analytical data at the time of analysis and reporting and through subsequent reviews of the raw data for non-conformances to the requirements of the analytical method. Laboratory personnel will make a systematic effort to identify outliers or errors before they report the data. Outliers that result from errors found during data verification will be identified and corrected. Outliers that cannot be attributed to errors in analysis, transcription, or calculation will be clearly identified in the case narrative section of the analytical data package.

10.0 DATA ANALYSIS AND REPORTING

Statistical analysis will compare groundwater data having greater than baseline concentrations once operations begin, as well as comparing down-gradient concentrations with up-gradient concentrations prior to operation, to ascertain if the sampling results are statistically significant. Similarly, statistical analyses will be used to compare soil and sediment data. All environmental monitoring data for surface water, groundwater and soils will be assessed relative to baseline per 40 CFR 264.94 (a) (I). Field verification of significant concentrations may require additional confirmation sampling to verify that the concentration of a specific parameter is valid.

LEAD proposes to use the EPA software ProUCL for analysis of variance (ANOVA) consistent with 40 CFR Part 264.97(h) ANOVA analyses. ProUCL provides numerous and varied statistical methods and graphical tools to address environmental sampling and statistical issues.

ProUCL can run environmental data sets with and without non-detect (ND) data samples. Calculating upper statistical limits is a primary function of the software and the graphical analyses offered includes probability plots, histograms, box plots, and line/trend plots. Results for statistical intervals have several options and relevant cautions. Other available tests that may apply to groundwater monitoring include single and two-sample hypotheses tests, ANOVA, regression, trend evaluation, outlier, and goodness-of-fit tests.

Environmental monitoring data, along with a narrative report and pertinent maps will be submitted to PADEP within 60 days after sample collection. The report to PADEP will include statistical treatment of monitoring data, all sampler field log/notes including weather conditions for day of sampling, monitoring well purge records, contour maps, laboratory data packages with all analytical data (including internal lab QA/QC data), chain of custody sheets and an EDD in a format consistent with the PADEP's LandLinks environmental database.

REFERENCES

- Code of Federal Regulations (CFR). 2016a. “Title 40 Protection of Environment, Part 264 Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities, Subpart F Releases from Solid Waste Management Units.” <http://www.ecfr.gov/cgi-bin/text-idx?SID=f253f25455f4fb6dd5cc40faf9f0cbaf&mc=true&node=sp40.26.264.f&rgn=div6>.
- CFR. 2016b. “Title 40 Protection of Environment, Part 261 Identification and Listing of Hazardous Waste.” http://www.ecfr.gov/cgi-bin/text-idx?SID=f253f25455f4fb6dd5cc40faf9f0cbaf&mc=true&tpl=/ecfrbrowse/Title40/40cfr261_main_02.tpl.
- Commonwealth of Pennsylvania, Department of Environmental Protection (PADEP). 2001. “*Groundwater Monitoring Guidance Manual*.” December 1.
- Mangi Environmental Group (Mangi). 2010. “Final Environmental Assessment For Ammonium Perchlorate Motor Destruction For Letterkenny Munitions Center.” March.
- The Pennsylvania Code (Pa. Code). 2016a. Chapter 16. Accessed April 2016 at <http://www.pacode.com/secure/data/025/chapter16/chap16toc.html>.
- Pa. Code. 2016b. Chapter 250. Accessed April 2016 at <http://www.pacode.com/secure/data/025/chapter250/chap250toc.html>.
- Redhorse Corporation & El Dorado Engineering, Inc. (Redhorse/EDE). 2014. “RCRA Part B Permit Modification Application for AP Rocket Motor Destruction Confined Burn Chamber at Letterkenny Army Depot.” February.
- Weston Solutions, Inc. (Weston). 2015. “Final Work Plan for the Drilling, Installation and Development of Four Monitoring Wells at the Ammonium Perchlorate Rocket Motor Destruction (ARMD) Facility at Letterkenny Army Depot.” Letter report submitted to PADEP December 2.
- Weston. 2003. “Depot-Wide Quality Assurance Project Plan for Environmental Services to Support Investigation, Removal, and Disposal Activities Performed by Contractors at the Letterkenny Army Depot.” September 1998, Revision 1 – October 2003.
- U.S. Environmental Protection Agency (EPA). 2016. “Table of Regulated Drinking Water Contaminants” and “Secondary Drinking Water Standards.” Accessed April 2016 at <http://water.epa.gov/drink/contaminants/index.cfm>. Last updated January 6.
- EPA. 2015. “Regional Screening Level (RSL) Summary Table (TR=1E-06, HQ=1) November 2015.” <https://www.epa.gov/risk/regional-screening-levels-rsls-generic-tables-november-2015>. November.

EPA. 2014. "Test Methods for Evaluating Solid Waste, Physical/Chemical Methods (SW-846), Update V." Office of Solid Waste and Emergency Response. Washington, DC. July.

EPA. 2006. "Guidance on Systematic Planning Using the Data Quality Objectives Process, EPA QA/G-4." Office of Environmental Information. Washington, DC. EPA/240/B-06/001. February.

EPA. 2000. "Guidance for the Data Quality Objectives Process, EPA QA/G-4." Office of Environmental Information. Washington, DC. EPA/600/R-96/055. August.

EPA. 1992. "Specifications and Guidance for Obtaining Contaminant-Free Sampling Containers." OSWER Directive No. 9240.0-05A. April.

APPENDIX A

VOLATILE ORGANIC COMPOUND LIST

Volatile Organic Compounds List

Acetone
Benzene
Bromodichloromethane
Bromoform
2-Butanone
Bromomethane
Carbon tetrachloride
Carbon disulfid
Chloroethane
Chlorobenzene
Chloroform
Chloromethane
Dibromochloromethane
1,1-Dichloroethane
1,2-Dichloroethane
1,1-Dichloroethene
1,2-Dichloroethene (total)
1,2-Dichloropropane
cis-1,3-Dichloropropene
trans-1,3-Dichloropropene
Ethylbenzene
2-Hexanone
Methyl ethyl ketone
4-Methyl-2-pentanone (MIBK)
Methylene chloride
Styrene
1,1,2,2-Tetrachloroethane
Tetrachloroethene
Toluene
1,1,1-Trichloroethane
1,1,2-Trichloroethane
Trichloroethene
Vinyl chloride
Xylenes (total)

ATTACHMENT A

**PADEP WELL INSTALLATION PLAN APPROVAL LETTER
&
WESTON SOLUTIONS, INC. WELL INSTALLATION PLAN LETTER REPORT**



pennsylvania
DEPARTMENT OF ENVIRONMENTAL
PROTECTION

March 1, 2016

Greg Epstein
Letterkenny Army Depot
Environmental Management Division
1 Overcash Avenue
Chambersburg, PA 17201-4150

Re: Letterkenny Army Depot ARMD Facility
Permit No. PA6213820503
Monitoring well location and construction
Green, Hamilton & Letterkenny Townships, Franklin County

Dear Mr. Epstein:

The Department has received and reviewed Weston Solutions' 12/2/2015 letter "Final Work Plan for the Drilling, Installation and Development of Four Monitoring Wells at the Ammonium Perchlorate Rocket Motor Destruction (ARMD) Facility at Letterkenny Army Depot. The Department approves the proposed well construction and the locations proposed in Letterkenny's "Site Plan" (drawing number SK-2073, attached).

Form HW-B Professional Certification – Hazardous Waste Facility Construction, Repair or Closure Activity, documenting the well installation/construction should be submitted to the Department, including both lithologic and construction logs of each well, as well as surveyed latitude/longitude and elevation data. Water well completion reports should also be submitted to the PA Bureau of Topographic and Geologic Survey (<http://www.dcnr.state.pa.us/topogeo/groundwater/wdindex/index.htm>).

Please contact the Department once drilling activities are scheduled so that Department personnel may attend. If you have any questions concerning the above, please feel free to contact me at 717-705-4926 or timlong@pa.gov.

Sincerely,

Timothy R. Long, P.G.
Licensed Professional Geologist
Waste Management Program

Attachment



Weston Solutions, Inc.
1400 Weston Way
P.O. Box 2653
West Chester, Pennsylvania 19380
610-701-3000 • Fax 610-701-3187
www.westonsolutions.com

The Trusted Integrator for Sustainable Solutions

2 December 2015

Commander
Letterkenny Army Depot
1 Overcash Avenue
AMLD-EN, Bldg 14
Chambersburg, PA 17201-4150

Attn: Greg Epstein

W.O. No. 14697.002.001.1010

Subject: Final Work Plan for the Drilling, Installation and Development of Four Monitoring Wells at the Ammonium Perchlorate Rocket Motor Destruction (ARMD) Facility at Letterkenny Army Depot

Dear Greg:

Weston Solutions, Inc. (WESTON®) is pleased to submit this Draft Work Plan for the drilling, installation and development of 4 groundwater monitoring wells at the Ammonium Perchlorate Rocket Motor Destruction (ARMD) Facility at Letterkenny Army Depot (LEAD).

Monitoring Well Drilling and Installation Procedures

In general, the monitoring wells should be installed following the procedures outlined in the United States Army Corps of Engineers (USACE) Manual EM-1110-1-4000, "Monitoring Well Design Installation and Documentation at Hazardous, Toxic and Radioactive Waste (HTRW) Sites", 1 November 1998. The overburden portion of each monitoring well will be drilled using a 10-inch diameter air rotary or air hammer bit and advanced to a minimum of 5 feet into competent bedrock. A 6-inch diameter steel casing will be installed and tremie-grouted to ground surface using a 4% bentonite/cement slurry. The grout and casing will be allowed to set for a minimum of 12 hours prior to drilling out the bottom hole portion of each well. The bottom hole portions of the wells will be drilled using a nominal 6-inch diameter bit to a total depth (TD) of approximately 80 feet below ground surface (BGS), or a minimum of 20 feet below either the depth of the 1st major water bearing zone (>0.5 gpm) or the static level of the well, whichever is deeper. The wells will be completed as 2-inch diameter PVC wells using 10-slot screen and riser with the screen length designed to straddle the primary water-bearing fractures within the well. The annular space of the well screen will be sandpacked using a medium grained, well sorted silica sand which will be placed from the base of the borehole to approximately 2-3 feet above the top of the well screen. A bailer will then be used to mechanically develop the sandpack of fines and to promote the settling of the sandpack prior to placing the bentonite seal. The depth to the top of the sandpack will be measured to verify that the sandpack after settling is still a minimum of 2 feet above the top of the well screen. A 2-foot thick bentonite pellet seal will then be placed above the sandpack and the remaining borehole will be tremie grouted to surface with a 4% bentonite/cement slurry. Figure 1 depicts a generalized well construction diagram for the proposed monitoring wells.

Throughout the drilling process, the borehole and all drill cuttings will be continuously screened for the presence of organic vapors using a calibrated flame ionization detector (FID). Continuously collected drill cuttings will also be washed, described and archived by a WESTON field geologist. Each well will be completed at the surface with either a 2-ft casing stickup or a flush-mounted well cover, as directed by the facility personnel. All surface completions will set in 2x2 ft concrete pads and 2-3 protective bollards, if deemed necessary by facility personnel. Following the drilling and installation activities, the top of the inner PVC casing and ground surface elevations (referenced to mean sea level), and northing and easting



coordinates of each well will be surveyed by a licensed Pennsylvania surveyor. All downhole drilling equipment and the undercarriage of the drill table will be decontaminated between well locations by steam cleaning with potable water.

Monitoring Well Development Procedures

Each of the wells will be developed following completion using stainless steel, submersible pumps. Throughout development, general water quality parameters will be measured using a YSI flow-through cell or equivalent. Pumping development of the wells will continue until 5-minute interval readings of the general water quality parameters stabilize according to the following criteria:

- | | |
|-------------------------|------------|
| ▪ pH: | ±0.1 units |
| ▪ Specific conductance: | ±10% |
| ▪ ORP: | ±10 mv |
| ▪ DO: | ±10% |
| ▪ Temperature: | ±0.1 °C |
| ▪ Turbidity | <50 NTU |

All well development activities will be recorded on well development field forms and will include (but will not be limited to) the times/dates of readings, static and pumping depths to water, flow rates and all general water quality measurements listed above. Completed lithologic logs, well construction diagrams and well development field forms will be provided to the Army upon completion of the wells, along with the surveyors report containing the surveyed well elevations and horizontal coordinates.

We appreciate the opportunity to provide continuing assistance to Letterkenny Army Depot. If you have any questions or comments regarding this Work Plan, or if we can assist you with any other matters, please call me at (610) 701-3489 or Paul Landry at (610) 701-3798.

Very truly yours,

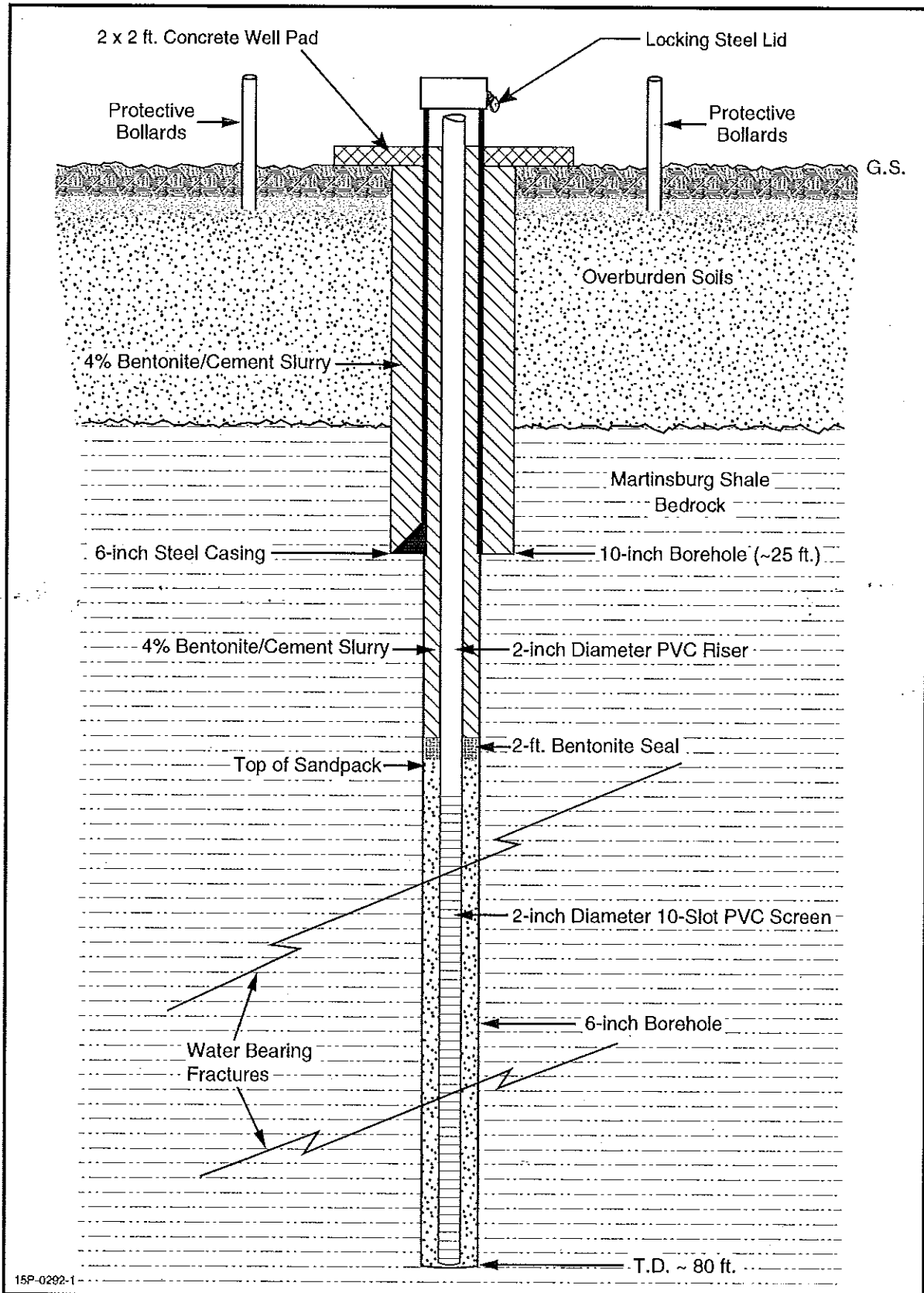
WESTON SOLUTIONS, INC.

A handwritten signature in black ink, appearing to read "Math Clapp", written over a horizontal line.

Mathew Clapp
Project Manager

Cc: J. Kohler (HF Lenz)
P. Landry (Weston)

FIGURES



**FIGURE 1 SCHEMATIC DIAGRAM OF SCREENED SHALLOW BEDROCK MONITORING WELL
LETTERKENNY ARMY DEPOT - CHAMBERSBURG, PA**

ATTACHMENT B

DEPOT-WIDE ANALYTICAL METHODS AND REPORTING LIMITS

Excerpt from the “Depot-Wide Quality Assurance Project Plan for Environmental Services to Support Investigation, Removal, and Disposal Activities Performed by Contractors at the Letterkenny Army Depot, September 1998, Revision 1 – October 2003” (Weston 2003).

Table 6-1

Analytical Methods and Reporting Limits

Parameter and Method	Reporting Limits ^a				
	EPA CLP Limits*		SW 846 Limits		
	Water OLM04.2/ OLM02.1 (µg/L)	Soil OLM04.2 (µg/kg)	Water 8260B/ LL8260B (µg/L)	Soil/Solids/Wipes ^a 8260B (µg/kg or µg/wipe)	Fish (µg/kg)
VOLATILES — EPA CLP SOW for Organic Analysis (OLM04.2/OLM02.1* or SW-846 Method 5030B (w)/5035 (s) [prep];/8260B & Low Level 8260 [analysis]					
Chloromethane	10/1.0	10	10/2.0	10/1,000 ^b	10
Bromomethane	10/1.0	10	10/1.0	10/1,000 ^b	10
Vinyl chloride	10/1.0	10	10/2.0	10/1,000 ^b	10
Chloroethane	10/1.0	10	10/2.0	10/1,000 ^b	10
Methylene chloride	10/2.0	10	5/2.0	5/500 ^b	5
Acetone	10/5.0	10	10/10	10/1,000 ^b	10
Carbon disulfide	10/1.0	10	5/1.0	5/500 ^b	5
1,1-Dichloroethene	10/1.0	10	5/1.0	5/500 ^b	5
1,1-Dichloroethane	10/1.0	10	5/1.0	5/500 ^b	5
1,2-Dichloroethene (total)	10/NA	10	5/1.0	5/500 ^b	5
Chloroform	10/1.0	10	5/1.0	5/500 ^b	5
1,2-Dichloroethane	10/1.0	10	5/1.0	5/500 ^b	5
2-Butanone	10/5.0	10	10/5.0	10/1,000 ^b	10
1,1,1-Trichloroethane	10/1.0	10	5/1.0	5/500 ^b	5
Carbon tetrachloride	10/1.0	10	5/1.0	5/500 ^b	5
Bromodichloromethane	10/1.0	10	5/1.0	5/500 ^b	5
1,2-Dichloropropane	10/1.0	10	5/1.0	5/500 ^b	5
Trichloroethene	10/1.0	10	5/1.0	5/500 ^b	5
Dibromochloromethane	10/1.0	10	5/1.0	5/500 ^b	5
1,1,2-Trichloroethane	10/1.0	10	5/1.0	5/500 ^b	5
Benzene	10/1.0	10	5/1.0	5/500 ^b	5
Bromoform	10/1.0	10	5/1.0	5/500 ^b	5
4-Methyl-2-pentanone	10/5.0	10	10/5.0	10/1,000 ^b	10
2-Hexanone	10/5.0	10	10/5.0	10/1,000 ^b	10
Tetrachloroethene	10/1.0	10	5/1.0	5/500 ^b	5
Toluene	10/1.0	10	5/1.0	5/500 ^b	5
1,1,2,2-Tetrachloroethane	10/1.0	10	5/1.0	5/500 ^b	5
Chlorobenzene	10/1.0	10	5/1.0	5/500 ^b	5
Ethylbenzene	10/1.0	10	5/1.0	5/500 ^b	5
Styrene	10/1.0	10	5/1.0	5/500 ^b	5
Xylenes (total)	10/1.0	10	5/1.0	5/500 ^b	5
Bromochloromethane	NA/1.0	NA	NA/NA	NA/NA	NA
Cyclohexane	10/NA	10	5/1.0	5/500 ^b	5
1,2-Dibromoethane(EDB)	10/1.0	10	NA/NA	NA/NA	NA
1,2-Dibromo-3-chloropropane(DBCP)	10/1.0	10	NA/NA	NA/NA	NA
1,2-Dichlorobenzene	10/1.0	10	NA/NA	NA/NA	NA
1,3-Dichlorobenzene	10/1.0	10	NA/NA	NA/NA	NA
1,4-Dichlorobenzene	10/1.0	10	NA/NA	NA/NA	NA
1,2-Dichloroethene (cis)	10/1.0	10	5/1.0	5/500 ^b	5
1,2-Dichloroethene (trans)	10/1.0	10	5/1.0	5/500 ^b	5

*Analyzed as per statement of work

Table 6-1

**Analytical Methods and Reporting Limits
(Continued)**

Parameter and Method	Reporting Limits ^a				
	EPA CLP Limits*		SW 846 Limits		
	Water OLM04.2/ OLM02.1 (µg/L)	Soil OLM04.2 (µg/kg)	Water 8260B/ LL8260B (µg/L)	Soil/Solids/Wipes ^a 8260B (µg/kg or µg/wipe)	Fish (µg/kg)
Dichlorodifluoromethane	10/NA	10	NA/NA		
1,3-Dichloropropene (cis)	10/1.0	10	5/1.0	5/500 ^b	5
1,3-Dichloropropene (trans)	10/1.0	10	5/1.0	5/500 ^b	5
Isopropylbenzene(Cumene)	10/NA	10	NA/NA	NA/NA	NA
Methyl Acetate	10/NA	10	NA/NA	NA/NA	NA
Methylcyclohexane	10/NA	10	NA/NA	NA/NA	NA
Methyl-tert-butyl-ether (MTBE)	10/NA	10	NA/NA	NA/NA	NA
1,2,4-Trichlorobenzene	10/1.0	10	NA/NA	NA/NA	NA
Trichlorofluoromethane	10/NA	10	NA/NA	NA/NA	NA
1,1,2-Trichloro-1,2,2- trifluoroethane(Freon 113)	10/NA	10	NA/NA	NA/NA	NA

Parameter and Method	Reporting Limits				
	EPA CLP Limits*		SW 846 Limits		
	Water OLM03.2 (ug/L)	Soil OLM03.2 (ug/Kg)	Water SW 846 (ug/L)	Soil/Solids/Wipes SW 846 (ug/Kg)	Fish SW 846 (ug/Kg)
BETX EPA CLP SOW for Organic Analysis (OLM03.2)* or Method, 5030B(w) 5035(s)/[prep];8021A]					
Benzene	10	10	5	5/500 ^b	5
Ethylbenzene	10	10	5	5/500 ^b	5
Toluene	10	10	5	5/500 ^b	5
Xylenes (total)	10	10	5	5/500 ^b	5

Parameter and Method	Reporting Limits				
	EPA CLP Limits*		SW 846 Limits		
	Water OLM04.2/ OLM02.1 (µg/L)	Soil OLM04.2 (µg/kg)	Water (µg/L)	Soil/Solids/Wipes ^a (µg/kg or µg/wipe)	Fish (µg/kg)
SEMIVOLATILES — EPA SOW for Organic Analysis (OLM04.2/OLM02.1)* orSW-846 Method 3520C (w)/3540B (s)/[prep]; 8270C (or 8310 for polynuclear aromatic hydrocarbons only) [analysis]					
Phenol	10/5	330	10	330	660
bis(2-chloroethyl) ether	10/5	330	10	330	660
2-Chlorophenol	10/5	330	10	330	660
1,3-Dichlorobenzene	NA/5	330	10	330	660
1,4-Dichlorobenzene	NA/5	330	10	330	660
1,2-Dichlorobenzene	NA/5	330	10	330	660
2-Methylphenol	10/5	330	10	330	660
2,2'-oxybis (1-Chloropropane)	10/5	330	10	330	660

*Analyzed as per statement of work

Table 6-1

**Analytical Methods and Reporting Limits
(Continued)**

Parameter and Method	Reporting Limits				
	EPA CLP Limits*		SW 846 Limits		
	Water OLM04.2/ OLM02.1 (µg/L)	Soil OLM04.2 (µg/kg)	Water (µg/L)	Soil/Solids/Wipes ^a (µg/kg or µg/wipe)	Fish (µg/kg)
4-Methylphenol	10/5	330	10	330	660
N-nitroso-di-n-propylamine	10/5	330	10	330	660
Hexachloroethane	10/5	330	10	330	660
Nitrobenzene	10/5	330	10	330	660
Isophorone	10/5	330	10	330	660
2-Nitrophenol	10/5	330	10	330	660
2,4-Dimethylphenol	10/5	330	10	330	660
bis(2-Chloroethoxy)methane	10/5	330	10	330	660
2,4-Dichlorophenol	10/5	330	10	330	660
1,2,4-Trichlorobenzene	NA/5	330	10	330	660
Napthalene	10/5	330	10	330	660
4-Chloroaniline	10/5	330	10	330	1,300
Hexachlorobutadiene	10/5	330	10	330	660
4-Chloro3-methylphenol	10/5	330	10	330	1,300
Hexachlorocyclopentadiene	10/5	330	50	1650	3300
2,4,6-Trichlorophenol	10/5	330	10	330	660
2,4,5-Trichlorophenol	25/20	830	10	330	660
2-Chloronaphthalene	10/5	330	10	330	660
2-Nitroaniline	25/20	830	50	1,650	3300
Dimethylphthalate	10/5	330	10	330	660
Acenaphthylene	10/5	330	10	330	660
2,6-Dinitrotoluene	10/5	330	10	330	660
3-Nitroaniline	25/20	830	50	1,650	3300
Acenaphthene	10/5	330	10	330	660
2,4-Dinitrophenol	25/20	830	50	1,650	3,300
Fluorene	10/5	330	10	330	660
4-Nitroaniline	25/20	830	50	1,650	3,300
4,6-Dinitro-2-methylphenol	25/20	830	50	1,650	3,300
N-nitrosodipheylamine	10/5	330	10	330	660
4-Bromophenyl-phenyl ether	10/5	330	10	330	660
Hexachlorobenzene	10/5	330	10	330	660
Pentachlorophenol	25/20	830	50	1,650	3,300
Phenanthrene	10/5	330	10	330	660
Anthracene	10/5	330	10	330	660
Carbazole	10/5	330	10	330	660
Di-n-butylphthalate	10/5	330	10	330	660
Fluoranthene	10/5	330	10	330	660
Pyrene	10/5	330	10	330	660
Butylbenzylphthalate	10/5	330	10	330	660
3,3'-Dichlorobenzidine	10/5	330	50	1650	3300
Benzo(a)anthracene	10/5	330	10	330	660
Chrysene	10/5	330	10	330	660

*Analyzed as per statement of work

Table 6-1

**Analytical Methods and Reporting Limits
(Continued)**

Parameter and Method	Reporting Limits				
	EPA CLP Limits*		SW 846 Limits		
	Water OLM04.2/ OLM02.1 (µg/L)	Soil OLM04.2 (µg/kg)	Water (µg/L)	Soil/Solids/Wipes ^a (µg/kg or µg/wipe)	Fish (µg/kg)
bis(2-ethylhexyl)phthalate	10/5	330	10	330	660
Di-n-octyl phthalate	10/5	330	10	330	660
Benzo(b)fluoranthene	10/5	330	10	330	660
Benzo(k)fluoranthene	10/5	330	10	330	660
Benzo(a)pyrene	10/5	330	10	330	660
Indeno(1,2,3-cd)pyrene	10/5	330	10	330	660
Dibenzo(a,h) anthracene	10/5	330	10	330	660
Benzo(g,h,i)perylene	10/5	330	10	330	660
Acetophenone	10/NA	330	NA	NA	NA
Benzaldehyde	10/NA	330	NA	NA	NA
Caprolactam	10/NA	330	NA	NA	NA
Atrazine	10/NA	330	NA	NA	NA
Biphenyl	10/NA	330	NA	NA	NA
4-Nitrophenol	25/20	830	50	1,650	3,300
Dibenzofuran	10/5	330	10	330	660
2,4-Dinitrotoluene	10/5	330	10	330	660
Diethylphthalate	10/5	330	10	330	660
4-Chlorophenyl-phenyl ether	10/5	330	10	330	660

Parameter and Method	Reporting Limits				
	EPA CLP Limits*		SW 846 Limits		
	Water OLM04.2 (µg/L)	Soil OLM04.2 (µg/kg)	Water (µg/L)	Soil/Solids/Wipes ^a (µg/kg or µg/wipe)	Fish (µg/kg)
PESTICIDES — EPA CLP SOW for Organic Analysis (OLM04.2.)* or SW-846 Method 3520C (w)/3550B (s)/3620 (s)/3540B (s) [prep]; 8081A [analysis]					
gamma-BHC (Lindane)	0.05/0.01	1.7	0.05	1.7	1.7
Heptachlor	0.05/0.01	1.7	0.05	1.7	1.7
Aldrin	0.05/0.01	1.7	0.05	1.7	1.7
Heptachlor epoxide	0.05/0.01	1.7	0.05	1.7	1.7
Endosulfan I	0.05/0.01	1.7	0.05	1.7	1.7
Dieldrin	0.10/0.02	3.3	0.05	1.7	1.7
4,4'-DDE	0.10/0.02	3.3	0.05	1.7	1.7
Endrin	0.10/0.02	3.3	0.05	1.7	1.7
Endosulfan II	0.10/0.02	3.3	0.05	1.7	1.7
4,4'-DDD	0.10/0.02	3.3	0.050	1.7	1.7
alpha-BHC	0.05/0.01	1.7	0.05	1.7	1.7
beta-BHC	0.05/0.01	1.7	0.05	1.7	1.7
delta-BHC	0.05/0.01	1.7	0.05	1.7	1.7
Endosulfan sulfate	0.10/0.02	3.3	0.050	1.7	1.7
4,4'-DDT	0.10/0.02	3.3	0.050	1.7	1.7

*Analyzed as per statement of work

Table 6-1

**Analytical Methods and Reporting Limits
(Continued)**

Parameter and Method	Reporting Limits				
	EPA CLP Limits*		SW 846 Limits		
	Water OLM04.2 (µg/L)	Soil OLM04.2 (µg/kg)	Water (µg/L)	Soil/Solids/Wipes ^a (µg/kg or µg/wipe)	Fish (µg/kg)
Methoxychlor	0.50/0.10	17.0	0.1	3.3	3.3
Endrin ketone	0.10/0.02	3.3	0.050	1.7	1.7
Endrin aldehyde	0.10/0.02	3.3	0.050	1.7	1.7
alpha-Chlordane	0.05/0.01	1.7	0.05	1.7	1.7
gamma-Chlordane	0.05/0.01	1.7	0.05	1.7	1.7
4,4'-DDT	0.10/0.02	3.3	0.050	1.7	1.7
Methoxychlor	0.50/0.10	17.0	0.1	3.3	3.3
Endrin ketone	0.10/0.02	3.3	0.050	1.7	1.7
Endrin aldehyde	0.10/0.02	3.3	0.050	1.7	1.7
alpha-Chlordane	0.05/0.01	1.7	0.05	1.7	1.7
gamma-Chlordane	0.05/0.01	1.7	0.05	1.7	1.7
Toxaphene	5.0/1.0	170	2.0	67	67
PCBs (AROCLORS) — EPA CLP SOW for Organic Analysis (OLM04.2)* or SW-846 Method 3520C (w)/3620 (s)/3550B(s)/3540B (s) [prep]; 8082A [prep]/SW-846 Method 6010 [prep]/1668 [analysis] for High Resolution (second limit shown)					
Aroclor 1016	0.20	33.0	1.0/0.01	33.0/1.0	33.0/1.01
Aroclor 1221	0.40	67.0	1.0/0.01	33.0/1.0	33.0/1.01
Aroclor 1232	0.20	33.0	1.0/0.01	33.0/1.0	33.0/1.01
Aroclor 1242	0.20	33.0	1.0/0.01	33.0/1.0	33.0/1.01
Aroclor 1248	0.20	33.0	1.0/0.01	33.0/1.0	33.0/1.01
Aroclor 1254	0.20	33.0	1.0/0.01	33.0/1.0	33.0/1.01
Aroclor 1260	0.20	33.0	1.0/0.01	33.0/1.0	33.0/1.01
HERBICIDES^c — SW-846 Method 8151A					
2,4-D	N/A	N/A	4.0	80	200
2,4-DB	N/A	N/A	4.0	80	200
2,4,5-TP (Silvex)	N/A	N/A	1.0	20	200
2,4,5-T	N/A	N/A	1.0	20	200
Dinoseb	N/A	N/A	0.6	12	200
Dalapon	N/A	N/A	2.0	40	200
MCPP	N/A	N/A	400	8000	200
Dicamba	N/A	N/A	2.0	40	200
Dichlorprap	N/A	N/A	4.0	80	200
MCPA	N/A	N/A	400	8000	200
ORGANIC PHOSPHOROUS PESTICIDES (OPPs)^c — SW-846 3520C (w)/3540B (s) [prep]; Method 8141A [analysis]					
Diazinon	N/A	N/A	1.0	33	134
Malathion	N/A	N/A	1.0	33	134
PCDDs/PCDFs - SW-846 Method 8290 or 1613 (high resolution)					
2,3,7,8-TCDD	N/A	N/A	0.005	1	TBD
2,3,7,8-TCDF	N/A	N/A	0.01	1	TBD
1,2,3,7,8-PeCDD	N/A	N/A	0.01	1	TBD
1,2,3,7,8-PeCDF	N/A	N/A	0.01	1	TBD
1,1,2,3,7,8-HxCDF	N/A	N/A	0.01	1	TBD

*Analyzed as per statement of work

Table 6-1

**Analytical Methods and Reporting Limits
(Continued)**

Parameter and Method	Reporting Limits				
	EPA CLP Limits*		SW 846 Limits		
	Water OLM04.2 (µg/L)	Soil OLM04.2 (µg/kg)	Water (µg/L)	Soil/Solids/Wipes ^a (µg/kg or µg/wipe)	Fish (µg/kg)
1,2,3,4,6,8-HpCDD	N/A	N/A	0.01	1	TBD
1,2,3,4,6,7,8-HpCDF	N/A	N/A	0.01	1	TBD

Parameter and Method	Reporting Limits				
	EPA CLP Limits		SW 846 Limits		
	Water ILM04.0 ICP/ICP Trace (µg/L)	Low Soil ILM04.0 ICP/ICP Trace (µg/kg)	Water (µg/L)	Soil/Solids/Wipes ^a (µg/kg or µg/wipe)	Fish (µg/kg)
METALS/INORGANICS — EPA SOW for Inorganic Analysis (ILM04.0)* or SW-846 Methods 3010A (w)/3020A (w)/3050a (s) [prep]; 6010B, 7471A, 7196A, 7000 series, or as specified below [analysis]					
Aluminum	200/NA	40,000/NA	200	20,000	20
Antimony	60/10	12,000/2,000	60	6,000	10
Arsenic	300/10	60,000/2,000	10	1,000	10
Barium	200/NA	40,000/NA	200	20,000	5
Beryllium	5/NA	1000/NA	5	500	1
Boron	200/NA	40,000/NA	N/A	N/A	N/A
Cadmium	5/NA	1000/NA	5	500	1
Calcium	5,000/NA	1,000,000/NA	5,000	500,000	10
Chromium (total)	10/NA	2,000/NA	10	1,000	2
Cobalt	50/NA	10,000/NA	50	5,000	2
Copper	25/NA	5,000/NA	25	2,500	2
Iron	100/NA	20,000/NA	100	10,000	3
Lead	150/3.0	30,000/600	3	300	5
Magnesium	5,000/NA	1,000,000/NA	5,000	500,000	20
Manganese	15/NA	3000/NA	15	1,500	1
Mercury	NA/NA	NA/NA	0.2	100	0.1
Nickel	40/NA	8,000/NA	40	4,000	2
Potassium	5,000/NA	1,000,000/NA	5,000	500,000	200
Selenium	250/5.0	50,000/1,000	5	500	10
Silver	10/NA	2,000/NA	10	1,000	1
Sodium	5,000/NA	1,000,000/NA	5,000	500,000	20
Thallium - 7841	NA/10	NA/2,000	10	1,000	50
Vanadium	50/NA	10,000/NA	50	5,000	1
Zinc	20/NA	4,000/NA	20	2,000	1
Cyanide - 9012A/9014	NA/NA	NA/NA	10	1,000	NA

*Analyzed as per statement of work

Table 6-1

**Analytical Methods and Reporting Limits
(Continued)**

Reporting Limits					
Parameter and Method	EPA CLP Limits*		SW 846 Limits		
	Water (µg/L)	Soil (µg/kg)	Water (mg/L)	Soil/Solids/Wipes ^a (µg/kg or µg/wipe)	Fish (µg/kg)
TCLP VOLATILES^c — SW-846 Method 1311/8260B					
Vinyl chloride	N/A	N/A	0.050	N/A	N/A
1,1-Dichloroethene	N/A	N/A	0.050	N/A	N/A
Chloroform	N/A	N/A	0.050	N/A	N/A
1,2-Dichloroethane	N/A	N/A	0.050	N/A	N/A
2-Butanone	N/A	N/A	0.050	N/A	N/A
Carbon tetrachloride	N/A	N/A	0.050	N/A	N/A
Trichloroethene	N/A	N/A	0.050	N/A	N/A
Benzene	N/A	N/A	0.050	N/A	N/A
Tetrachloroethene	N/A	N/A	0.050	N/A	N/A
Chlorobenzene	N/A	N/A	0.050	N/A	N/A
TCLP SEMIVOLATILES^c — SW-846 Method 1311/8270C					
Pyridine	N/A	N/A	0.10	N/A	N/A
1,4-Dichlorobenzene	N/A	N/A	0.050	N/A	N/A
Cresols(Total)	N/A	N/A	0.050	N/A	N/A
Hexachloroethane	N/A	N/A	0.050	N/A	N/A
Nitrobenzene	N/A	N/A	0.050	N/A	N/A
Hexachlorobutadiene	N/A	N/A	0.050	N/A	N/A
2,4,6-Trichlorophenol	N/A	N/A	0.050	N/A	N/A
2,4,5-Trichlorophenol	N/A	N/A	0.050	N/A	N/A
2,4-Dinitrotoluene	N/A	N/A	0.050	N/A	N/A
Hexachlorobenzene	N/A	N/A	0.050	N/A	N/A
Pentachlorophenol	N/A	N/A	0.250	N/A	N/A
TCLP METALS^c — SW-846 Methods 1311/6010A, 7471A, 7196A, or 7000 series					
Silver	N/A	N/A	0.500	N/A	N/A
Arsenic	N/A	N/A	0.500	N/A	N/A
Barium	N/A	N/A	10.0	N/A	N/A
Cadmium	N/A	N/A	0.100	N/A	N/A
Chromium	N/A	N/A	0.500	N/A	N/A
Copper	N/A	N/A	0.025	N/A	N/A
Mercury	N/A	N/A	0.0002	N/A	N/A
Nickel	N/A	N/A	0.040	N/A	N/A
Lead	N/A	N/A	0.500	N/A	N/A
Selenium	N/A	N/A	0.250	N/A	N/A
Zinc	N/A	N/A	0.020	N/A	N/A
TCLP PESTICIDES/HERBICIDES^c — SW-846 Methods 1311/8081A, or 8151A					
Heptachlor	N/A	N/A	0.0005	N/A	N/A
alpha-Chlordane(technical)	N/A	N/A	0.005	N/A	N/A
gamma Chlordane	N/A	N/A	0.0005	N/A	N/A
gamma-BHC (Lindane)	N/A	N/A	0.0005	N/A	N/A
Endrin	N/A	N/A	0.0005	N/A	N/A
Methoxychlor	N/A	N/A	0.001	N/A	N/A
Toxaphene	N/A	N/A	0.02	N/A	N/A

*Analyzed as per statement of work

Table 6-1

**Analytical Methods and Reporting Limits
(Continued)**

Reporting Limits					
Parameter and Method	EPA CLP Limits*		SW 846 Limits		
	Water (µg/L)	Soil (µg/kg)	Water (mg/L)	Soil/Solids/Wipes ^a (µg/kg or µg/wipe)	Fish (µg/kg)
Heptachlor epoxide	N/A	N/A	0.0005	N/A	N/A
2,4-D	N/A	N/A	0.04	N/A	N/A
2,4,5-TP (Silvex)	N/A	N/A	0.01	N/A	N/A
EXPLOSIVES – SW 846 Method 8330					
HMX	NA	NA	0.1	250	NA
RDX	NA	NA	0.1	250	NA
1,3,5-trinitrobenzene	NA	NA	0.1	250	NA
1,3-dinitrobenzene	NA	NA	0.1	250	NA
Tetryl	NA	NA	0.1	250	NA
Nitrobenzene	NA	NA	0.1	250	NA
2,4,6-trinitrotoluene	NA	NA	0.1	250	NA
4-Am-dinitrotoluene	NA	NA	0.1	250	NA
2-Am-dinitrotoluene	NA	NA	0.1	250	NA
2,4-dinitrotoluene	NA	NA	0.1	250	NA
2,6-dinitrotoluene	NA	NA	0.1	250	NA
2-nitrotoluene	NA	NA	0.5	250	NA
3-nitrotoluene	NA	NA	0.5	250	NA
4-nitrotoluene	NA	NA	0.5	250	NA
PERCHLORATE – Method 314.0					
Perchlorate	NA	NA	4.0	50	NA
OTHER PARAMETERS^c					
TPH ^c (418.1 modified or SW-846 9071A/9073)	N/A	N/A	250	30,000	N/A
TPH gasoline range organics (SW-846 8015 modified) ^c	N/A	N/A	30,000	30,000	N/A
TPH diesel range organics ^c (SW-846 8015 modified)	N/A	N/A	1,000	10,000	N/A
Tannin (method to be determined) ^c	N/A	N/A	TBD	TBD	N/A
Chromium, hexavalent ^c SW-846 7196/7199	N/A	N/A	0.01 mg/L	0.40mg/Kg	N/A
Asbestos ^c EPA 600/M4-82/020	N/A	N/A	1%	1%	N/A
Oil and Grease ^c (418.1 or SW-846 9070)	N/A	N/A	5 mg/L (grav); 1.0mg/L (IR)	33mg/Kg (IR)	N/A
COD ^c (EPA 410.2/410.4)	N/A	N/A	10 mg/L	100mg/Kg	N/A
Ammonia ^c (EPA 350.2 or 350.1 modified)	N/A	N/A	0.10mg/L	5.0mg/Kg	N/A
Ignitability ^c (SW-846 1010)	N/A	N/A	± 1°C (<200° F)	yes/no (<200° F)	N/A
pH ^c (SW-846 9040B/9045C or EPA 150.1)	N/A	N/A	± 0.1 units	± 0.1 units	N/A
TOC ^c (modified Lloyd Kahn,	N/A	N/A	1.0 mg/L	10mg/Kg	N/A

*Analyzed as per statement of work

Table 6-1

**Analytical Methods and Reporting Limits
(Continued)**

Reporting Limits					
Parameter and Method	EPA CLP Limits*		SW 846 Limits		
	Water (µg/L)	Soil (µg/kg)	Water (mg/L)	Soil/Solids/Wipes ^a (µg/kg or µg/wipe)	Fish (µg/kg)
SW-846 9060 or EPA 415.1)					
Total Sulfide ^c (SW-846 9030 or 376.1-)	N/A	N/A	1.0 mg/L	5.0 mg/kg	N/A
Reactive Sulfide ^c (Ch. 7.3 of SW-856 or SW-846 9030)	N/A	N/A	10	10mg/Kg	N/A
Reactive Cyanide ^c (Ch. 7.3 of SW-846 or SW-846 9010)	N/A	N/A	0.3	0.3mg/Kg	N/A
Paint Filter (SW846 9095) ^c	N/A	N/A	pass/fail	N/A	N/A
% Solids (ASTM D2216) ^c	N/A	N/A	N/A	0.1%	0.1%
% Volatile Solids (ASTM D2216) ^c	N/A	N/A	N/A	0.1%	0.1%
% Moisture ^c (ASTM D2216)	N/A	N/A	N/A	0.1%	0.1%
ASTM Leach (D3987) ^c	N/A	N/A	N/A	N/A	N/A
Total Dissolved and suspended solids (TDS/TSS) (EPA 160.1/160.2) ^c	N/A	N/A	10 mg/L(TDS) 4mg/L(TSS)	N/A	N/A
Biological oxygen demand (BOD ₅) (EPA 405.1) ^c	N/A	N/A	2.0 mg/L	N/A	N/A
Chloride (EPA 300) ^c	N/A	N/A	1.0ppm	N/A	N/A
Sulfate (EPA 375) ^c	N/A	N/A	5 mg/L	N/A	N/A
Nitrite/Nitrate (EPA 300) ^c	N/A	N/A	0.1ppm	N/A	N/A
Hardness (EPA 130.1) ^c	N/A	N/A	3 mg/L	N/A	N/A
Alkalinity (SW846 2320B) ^c	N/A	N/A	5.0mg/L	N/A	N/A
BTU content (ASTM D240) ^c	N/A	N/A	N/A	1 unit	N/A
Gross alpha/beta (Gamma Spectroscopy method LANL ER-130 mod., wipes or air) ^c	N/A	N/A	N/A	Air: 2x10 ⁻¹³ µCi/mL ^d Wipes ^e : 10/20 dpm/100 cm ²	N/A
Tritium (Liquid Scintillation Counting) ^c , water or wipes	N/A	N/A	700 pCi/L	100 dpm/100 cm ²	N/A
Sieve/Grain Size (ASTM D422) ^g	N/A	N/A	N/A	0.1% (per particle size)	N/A
Liquid & Plastic Limits (ASTM D4318) ^g	N/A	N/A	N/A	0.1 psi	N/A
Specific Gravity (ASTM D854) ^g	N/A	N/A	N/A	0.01 units	N/A
Moisture, Ash and Organic Content (ASTM D2974) ^g	N/A	N/A	N/A	0.1%	N/A
Permeability (ASTM D2434) ^g	N/A	N/A	N/A	0.01 darcys	N/A
Density (ASTM D2937) ^g	N/A	N/A	N/A	0.1 g/cc	N/A
Metals in Air (NIOSH or MM-29 or 7300)	N/A	N/A	N/A	Dependent on volume of air ^h	N/A
Particulates in Air (EPA5 or PM-10) ^g	N/A	N/A	N/A	1 particle per mL	N/A

*Analyzed as per statement of work

Table 6-1

**Analytical Methods and Reporting Limits
(Continued)**

Reporting Limits					
Parameter and Method	EPA CLP Limits*		SW 846 Limits		
	Water (µg/L)	Soil (µg/kg)	Water (mg/L)	Soil/Solids/Wipes ^a (µg/kg or µg/wipe)	Fish (µg/kg)
Asbestos in Air ^g	N/A	N/A	N/A	0.1%	N/A
Total Lipids ^c (EPA Fish Monitoring Guidance Document), see Appendix B	N/A	N/A	N/A	N/A	0.1%

^a Units are shown at top of column or as otherwise shown in each parameter listing.

^b Second number shown is reporting limit for methanol preservation under Method 5035.

^c Not a SW-846 and/or CLP parameter.

^d Reporting limit is dependent on volume of air sample; is it estimated that a minimum of 1x10⁶ mL of air is needed to achieve this RL.

^e 10 dpm/100 cm² for gross alpha and 20 dpm/100cm² for gross beta.

^g Reporting Limits are as per the test method.

^h Reporting Limit is compound specific and is dependent on volume of air sample as specified in the test method. This information will be provided in the Technical Plan or Site-Specific QAPP Checklist.

BOD = biological oxygen demand

cm² = Square centimeters.

dpm = Disintegrations per minute.

g/cc = Grams per cubic centimeter.

N/A = Not applicable.

µg/g = Micrograms per gram.

µg/kg = Micrograms per kilogram.

µg/L = Micrograms per liter.

mg/kg = Milligrams per kilogram.

mg/L = Milligrams per liter.

pCi = Pico curies.

psi = Pounds per square inch.

SW = Test Methods for Evaluating Solid Waste, Third Edition (Update II), September 1994 and December 1997.

TBD = To be determined and included in Technical Plan or Site Specific QAPP Checklist.

µCi = Micro curies.

*Analyzed as per statement of work